PROVEIT RESEARCH CONTINUATION

Bernoulli Cancellation, Harmonic Gaps, and Spectral Faces

All-power trigamma moments, a completed reconstruction family, and identities for Gamma and Stieltjes functions

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Abstract

We continue several precise research questions in the ProveIt manuscript and its five incoming reports of 11 October 2026. A single Bernoulli polynomial identity cancels the finite-cutoff boundary terms for every power of the trigamma tail. It gives an anchored analytic generator, a finite multiple-zeta coordinate space at each power, a uniform extension to all repeated even-order Hurwitz tails, and explicit fifth- and sixth-power ordinary convergent sums. We determine all Laurent coefficients on the complementary resonant lattice, including the exact local convergence radius and Stieltjes normalization. A separate exact certificate at the previously uncovered index completes the smaller-order-ten optimal-prefix reconstruction family. For relative multiple-Hurwitz interpolants we prove closure under finite decorations and obtain several-marker identities involving log-Gamma products and normalized Stieltjes primitives. Finally, an explicit subtraction on every face of a Dirichlet simplex expresses all higher spectral defects through lower one-mass derivatives; the full zero-jet correction is a finite rational function and vanishes for mean-zero periodic weights. Proofs, exact algebraic certificates, numerical diagnostics, a source audit, and precise further questions are supplied. Statements of rank refer to specified rational coordinate systems; no numerical period independence or unproved Gaussian ladder identity is asserted.

Source baseline. ProveIt commit `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`.

Status. Research article with complete stated proof chains and reproducible finite checks; prepared for subsequent repository review and integration.

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1 Scope, results, and the source questions

The source corpus is already extensive. It contains general harmonic interpolation, centered tail moments, Gamma and Stieltjes derivative ladders, finite spectral-subtraction algorithms, and a careful record of rejected or still-conjectural period relations. A continuation should therefore do more than rederive a familiar Euler sum or enlarge a numerical search. We select explicit unresolved questions, identify their inherited inputs, and supply complete new proof steps.

The baseline is commit `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`, including the canonical manuscript [1], the harmonic-gap continuation [9], and all five incoming archives dated 11 October 2026. In particular the quartic reconstruction report [2] already asks for fifth- and higher-power cancellation, the complementary resonant parity, and completion of the order-ten reconstruction family. The higher-defect report [10] asks for explicit simplex-face subtractions and their odd-weight extension. These are concrete targets addressed here.

1.1 The main new statements

The first contribution is an all-power mechanism. If

$$A_n = \psi'(n+1) = \zeta(2) - H_n^{(2)}, \qquad x_n = n + \tfrac{1}{2},$$

the regular centered moments of A_n^d have an anchored generating function satisfying one triangular recurrence for every d. Its finite boundary cancellation is the parity of

$$u^{2d} B_d\!\left(\tfrac{1}{2} + \frac{E(u)}{u}\right), \qquad E(u) = \sum_{j\ge 0} B_{2j} u^{2j}.$$

This polynomial identity explains the lower-power examples and proves the fixed finite-span conclusion without extrapolating from a finite table. The resulting fifth-power coordinate system has eight formal coordinates, and the sixth-power system has eleven. Exact rational determinants certify the full ranks of the displayed coefficient rows. An especially compact consequence is

$$\begin{aligned}\sum_{n=0}^{\infty}\Big[(195x_n + 4056x_n^3 + 11568x_n^5 + 5376x_n^7)A_n^5 - 5376x_n^2 - 9328\Big] \\ = -748\zeta(2) + 7884\zeta(3) - 5380\zeta(4) + 171\zeta(6) - 748.\end{aligned} \tag{1.1}$$

The entire bracket is one summand. Its series is ordinary and absolutely convergent; splitting off its divergent polynomial pieces would change the meaning. The proof is finite summation and elimination, not integer relation recognition.

The same mechanism works for every repeated even-order Hurwitz tail $\zeta(2q, n+1)^d$, with an explicit finite span of weights at most $2qd - 1$ (Theorem 2.5). A short consequence, with $T_n = \psi^{(3)}(n+1)/6$, is

$$\sum_{n\ge 0}\big((4x_n^3 + x_n)T_n^3 - 3T_n^2\big) = -\frac{9}{4}\zeta(8).$$

The underlying cubic cutoff identity also gives a complex-order family for $\Re p > 3/2$. At the critical real order $p = 3/2$, the ordinary sum acquires the exact additional endpoint contribution 8; see Corollary 2.7.

The same spectral family has actual poles at $1-d-2r$. We give all their Laurent coefficients as ordinary logarithmic sums plus a finite Hurwitz and generalized-Stieltjes correction. Removing the pole gives a Taylor series of exact radius two. This answers the normalization part of

the resonant question and supplies an analytic local generator, while leaving open a uniform arithmetic reduction as the resonance index increases.

The second contribution completes a previously partial theorem. For each even larger order $2b > 10$, the first $2b$ regular centered moments of $\zeta(10, n+1)\zeta(2b, n+1)$ have the optimal formal coordinate rank $2b$. Two existing reports cover complementary ranges: a congruence covers all but one residue class, and an eventual theorem covers $b \geq 1920$. Their only uncovered index is $b = 962$. A five-by-five matrix modulo 2^{20} proves nonvanishing there. We replay the inherited finite polynomial identities and give an explicit rigorous large-order comparison, so the completed family has no unchecked numerical case. The additional parameter pair relative to the union of the two reports is exactly $(10, 1924)$.

The third contribution treats several transverse marks in relative multiple-Hurwitz interpolation. Its first useful case is

$$\partial_u \partial_v \mathcal{H}_{a,b}(s,u,v,t)|_{u=v=0} = \frac{1}{2}\mathcal{E}^{s,t}_{a,b}\left[(\log\Gamma(z+1) - \log\Gamma(y))^2 - 2\int_y^{z+1} \gamma_1(x)\,dx\right]. \tag{1.2}$$

Here $\mathcal{H}$ is the prescribed relative quotient and $\mathcal{E}$ is an independently defined decorated double quotient; their full definitions and ordinary-series realizations appear in Section 6. The identity holds for arbitrary independent spectral arguments and endpoints in the right half-plane. Its extension uses a finite set-partition polynomial. Identical marked clusters give individual ordered identities; different marks give their complete permutation-symmetric sum. The analytic proof establishes decoration closure before using the partition identity.

The fourth contribution resolves the face-subtraction problem for higher spectral defects. For a polynomial weight, any number of nonnegative masses, and every derivative order, we give an exact finite sum of ordinary integrals over simplex faces. A k-vertex face first appears at derivative order k, after the zero-jet part is separated. That zero-jet part is an explicit rational function. It is absent for even polynomial weights and, more generally, for every polynomial weight multiplied by a mean-zero periodic sequence. The highest faces at each derivative order require only the first one-mass derivative, which for constant weight is an ordinary log-Gamma expression. The subtraction also gives a closed identity at every face size:

$$\int_{\Sigma_S} \mathcal{H}_S(B \mapsto B^2; t)\frac{dt}{\prod_{i\in S} t_i} = (-1)^{|S|-1}(|S|-2)! \sum_{\{i,j\}\subseteq S} (B_i - B_j)^2. \tag{1.3}$$

The simplex normalization and face operator are specified in Section 7; the integral has no hidden finite part.

1.2 Claim boundaries and relation to earlier work

Several ingredients are classical and are used with attribution. Hoffman's ordering of the minimum index gives products of zeta tails as multiple zeta values [5]. His symmetric-sum theorem supplies the partition coefficients [7, 8]. Ihara–Nakamura–Yamamoto supply the bare entire interpolation and its harmonic product [6]. The scalar Dirichlet mixture is a classical Gamma-integral identity, already present in the source; the literature on logarithmic Dirichlet averages includes [11, 12]. These facts are inputs, not claimed discoveries of the present article.

The new content is the all-power boundary mechanism and its concrete consequences; the missing exact reconstruction certificate and completed family; the local analytic realization of multiple decorations and marks; and the explicit face decomposition with its rational correction and all-order filtration. Novelty is assessed relative to the inspected repository and the cited primary literature. An exhaustive priority claim over the entire literature is not made.

The manuscript's Gaussian S_6 and revised S_8 period identities remain open. The rejected historical S_8 vector is a different object and retains its rejected status. No determinant of

a formal coordinate matrix proves arithmetic independence of evaluated periods. Likewise a representation by polylogarithms of bounded depth is an upper bound in that representation, not a proof of minimal depth. These distinctions are part of the theorem statements, not qualifications added only after the calculations.

1.3 Conventions and proof architecture

We use $H_n^{(r)} = \sum_{j=1}^{n} j^{-r}$, with $H_0^{(r)} = 0$; Bernoulli numbers have $B_1 = -1/2$; and multiple zeta values are ordered as

$$\zeta(s_1, \dots, s_\ell) = \sum_{n_1 > \cdots > n_\ell \geq 1} n_1^{-s_1} \cdots n_\ell^{-s_\ell}.$$

The prime on a zeta function denotes differentiation in its spectral argument unless explicitly stated otherwise. The generalized Stieltjes constants satisfy

$$\zeta(1+\varepsilon, a) = \frac{1}{\varepsilon} + \sum_{q \geq 0} \frac{(-1)^q \gamma_q(a)}{q!} \varepsilon^q.$$

Section-specific letters are reset explicitly: $F_d(s)$ denotes a tail spectrum, whereas $F_j(B)$ in the spectral-face section denotes a one-mass derivative. Every antiderivative has its endpoint fixed.

The arguments use three distinct kinds of evidence. Finite identities are proved algebraically and supplied with rational or modular replay. Analytic statements include convergence domains, continuation from an open domain, and endpoint normalization. High-precision evaluations then check independent representations and signs. Numerical agreement is not used as a substitute for either of the first two steps. Section 8 records what was audited and what the artifact checks establish; Section 9 lists exact next targets.

2 A Bernoulli cancellation law for every trigamma power

The quartic report asks for a cancellation mechanism at the fifth and higher powers, a fixed arithmetic coordinate space at each power, and a generator with its constants specified [2, Research question R4]. We give one recurrence for all powers. It explains the earlier quadratic, cubic, and quartic cancellations and yields new fifth- and sixth-power formulas. The essential ingredient is a Bernoulli polynomial, rather than a new summation theorem for each successive power.

2.1 Regular moments as ordinary convergent sums

Throughout this section let

$$A_n = \psi'(n+1) = \zeta(2) - H_n^{(2)}, \qquad x_n = n + \tfrac{1}{2}, \qquad n \geq 0. \tag{2.1}$$

Bernoulli numbers have the convention $B_1 = -1/2$. For $d \geq 1$ define

$$h_{d,j} = \sum_{j_1 + \cdots + j_d = j} \prod_{\nu=1}^{d} B_{2j_\nu}(1/2), \qquad j_\nu \geq 0. \tag{2.2}$$

In particular $h_{d,0} = 1$ and $h_{d,1} = -d/12$. The centered Hurwitz expansion [4, §25.11] gives

$$A_n^d \sim \sum_{j \geq 0} h_{d,j} x_n^{-d-2j}. \tag{2.3}$$

This is an asymptotic expansion; it is not a convergent Laurent series at a fixed finite n.

Proposition 2.1 (Ordinary realization of the regular parity)**.** *The function*

$$F_d(s) = \sum_{n\geq 0} A_n^d x_n^{-s}, \qquad \Re s > 1-d,$$

has a meromorphic continuation to $\mathbb{C}$*. Its poles are exactly* $1-d-2j$*,* $j \geq 0$*, and their residues are* $h_{d,j}$*. For every* $m \geq 0$ *with* $m \equiv d \pmod 2$*, the value* $M_{d,m} = F_d(-m)$ *is regular and is the absolutely convergent sum*

$$M_{d,m} = \sum_{n\geq 0} \left(x_n^m A_n^d - \sum_{0\leq j\leq (m-d)/2} h_{d,j} x_n^{m-d-2j} \right). \tag{2.4}$$

An inner sum with negative upper limit is empty.

Proof. Subtract a finite number of terms of (2.3) and add the corresponding functions $h_{d,j}\zeta(s+d+2j, 1/2)$. Increasing the subtraction order gives normally convergent remainders in successively larger half-planes. The meromorphic continuations agree on overlaps, since they agree with the defining series in its initial half-plane. The signs of $B_{2j}(1/2)$ are $(-1)^j$, including $j=0$. Every summand in (2.2) therefore has sign $(-1)^j$; hence $h_{d,j} \neq 0$ and the listed poles are actual.

At $s=-m$ the terms subtracted in (2.4) contribute $h_{d,j}\zeta(d+2j-m, 1/2) = 0$. Indeed their Hurwitz arguments are nonpositive even integers, and $\zeta(-2q, 1/2) = 0$ for $q \geq 0$. The next summand is $O(x_n^{-2})$, or decays faster when no subtraction is needed. This proves both the ordinary convergence and the equality. □

Thus the word "moment" below always refers to the explicit convergent expression (2.4). The meromorphic family is a proof and differentiation tool; no assignment of a value to an unspecified divergent sum is required.

2.2 Finite-cutoff coordinates

For a finite-cutoff asymptotic expansion, let CT_N denote the coefficient of $N^0(\log N)^0$. Put

$$\begin{aligned} T_{e,q} &= \mathrm{CT}_N \sum_{n=1}^{N} n^q A_n^e, \qquad b_e(k) = [u^k]\mathcal{A}(u)^e, \\ \mathcal{A}(u) &= \sum_{a\geq 1} B_{a-1} u^a, \\ C_{d,k} &= b_d(k) + \sum_{j=1}^{d} \binom{d}{j} T_{d-j,k-2j}. \end{aligned} \tag{2.5}$$

Here $b_0(k) = \delta_{k,0}$, and a coefficient with negative index is zero. The series $\mathcal{A}$ is formal. These constant coefficients exist: repeated finite summation by parts reduces their existence to ordinary power sums, harmonic numbers, and convergent multiple zeta values. A constructive reduction is given below.

The exact recurrence $A_{n-1} = A_n + n^{-2}$ shows that $C_{d,k}$ is the constant coefficient of

$$\sum_{n=1}^{N} n^k (A_{n-1}^d - A_n^d) + N^k A_N^d.$$

Define the centered Faulhaber polynomial

$$P_m(N) = \sum_{n=0}^{N-1} (n+1/2)^m = \frac{B_{m+1}(N+1/2) - B_{m+1}(1/2)}{m+1}. \tag{2.6}$$

Finite summation by parts gives

$$\sum_{n=0}^{N-1} x_n^m A_n^d = P_m(N)A_N^d + \sum_{n=1}^{N} P_m(n)(A_{n-1}^d - A_n^d). \tag{2.7}$$

Every subtracted even-power sum in (2.4) is an odd polynomial in N, so it has zero constant coefficient. Consequently, if $P_m(N) = \sum_{k\geq 1} p_{m,k}N^k$, then

$$M_{d,m} = \sum_{k\geq 1} p_{m,k}C_{d,k}. \tag{2.8}$$

The parity in (2.6) is useful. For even m the polynomial is odd; for odd m it is even with zero constant term. Formally define

$$G_d(t) = \sum_{\substack{m\geq 0 \\ m\equiv d \pmod 2}} M_{d,m}\frac{t^m}{m!}, \qquad H_d(t) = 2\sinh(t/2)G_d(t). \tag{2.9}$$

The finite polynomial identity (2.8) is equivalent to

$$H_d(t) = \sum_{\substack{k\geq 1 \\ k\equiv d+1 \pmod 2}} C_{d,k}\frac{t^k}{k!}. \tag{2.10}$$

For example, summing (2.6) over even m gives $\sinh(Nt)/(2\sinh(t/2))$; summing over odd m gives $(\cosh(Nt)-1)/(2\sinh(t/2))$. These two identities prove (2.10) coefficient by coefficient without assuming convergence of a moment generating series in advance.

2.3 The universal boundary cancellation

Lemma 2.2 (A finite Bernoulli difference identity)**.** *For every integer $d \geq 1$ and indeterminates a, h,*

$$\begin{aligned}(a+h)^d - a^d = \frac{dh}{2}\big((a+h)^{d-1} + a^{d-1}\big) \\ - \sum_{j=1}^{\lfloor (d-1)/2\rfloor} \binom{d}{2j} B_{2j}h^{2j}\big((a+h)^{d-2j} - a^{d-2j}\big).\end{aligned} \tag{2.11}$$

Moreover, whenever $k \equiv d+1 \pmod 2$,

$$b_d(k) + \frac{d}{2}b_{d-1}(k-2) + \sum_{j=1}^{\lfloor d/2\rfloor} \binom{d}{2j} B_{2j}b_{d-2j}(k-4j) = 0. \tag{2.12}$$

Proof. The first identity is the terminating Euler–Maclaurin identity for a polynomial on one interval. It also follows by comparing coefficients in the Bernoulli generating function, so its use here is purely finite algebra. A term with $d - 2j = 0$ would be a difference of constants and is omitted.

For the second identity write

$$E(u) = \sum_{j\geq 0} B_{2j}u^{2j}, \qquad \mathcal{A}(u) = uE(u) - u^2/2.$$

The expression on the left of (2.12) is the coefficient of u^k in

$$\begin{aligned}\mathcal{A}(u)^d + \frac{d}{2}u^2\mathcal{A}(u)^{d-1} + \sum_{j\geq 1}\binom{d}{2j}B_{2j}u^{4j}\mathcal{A}(u)^{d-2j} \\ = u^{2d}B_d\left(\frac{1}{2}+\frac{E(u)}{u}\right).\end{aligned} \tag{2.13}$$

The Bernoulli reflection law $B_d(1-z) = (-1)^d B_d(z)$ implies that the right side has parity d as a formal series in u. Its coefficient at the opposite parity is zero. Although the argument of B_d contains u^{-1}, the displayed product has no negative powers: the polynomial has degree d. This establishes the cancellation. □

Identity (2.13) supplies a single reason for all the boundary terms to cancel. The separate quartic cancellation is one low-degree consequence.

Theorem 2.3 (All-power triangular recurrence)**.** *Set $G_0 = 0$. For $d \geq 1$ and $k \geq 2$ with $k \equiv d+1 \pmod 2$,*

$$\boxed{\begin{aligned}C_{d,k} = d(k-2)![t^{k-2}](\cosh(t/2)G_{d-1}(t)) - \frac{d}{2}\delta_{k,2}\zeta(2)^{d-1} \\ - \sum_{j=1}^{\lfloor (d-1)/2\rfloor}\binom{d}{2j}B_{2j}C_{d-2j,k-4j}.\end{aligned}} \tag{2.14}$$

Every coordinate on the right has smaller tail power.

Proof. Apply (2.11) with $a = A_n$ and $h = n^{-2}$, multiply by n^k, and sum only up to a finite N. For $m = k-2 \geq 0$, changing the index in one of the two symmetric sums gives the exact relation

$$\begin{aligned}&\sum_{n=1}^{N} n^m(A_{n-1}^{d-1} + A_n^{d-1}) \\ &\quad = \sum_{n=0}^{N-1}\big((n+1)^m + n^m\big)A_n^{d-1} - \delta_{m,0}\zeta(2)^{d-1} + N^m A_N^{d-1}.\end{aligned}$$

The polynomial $(x+1/2)^m + (x-1/2)^m$ contains only powers of parity $m \equiv d-1 \pmod 2$. Taking its constant coefficient therefore uses exactly the regular moments of power $d-1$. Its contribution is $2m![t^m](\cosh(t/2)G_{d-1})$. For $d = 1$, these polynomial sums have zero constant coefficient, which is the convention $G_0 = 0$.

The differences of the lower powers in (2.11) give $C_{d-2j,k-4j} - b_{d-2j}(k-4j)$. The remaining boundary terms are precisely (2.12); their sum is zero. Taking constant coefficients after these finite identities have been combined proves (2.14). In particular the exceptional $k = 2$ term comes from the missing $n = 0$ endpoint and cannot be discarded. □

2.4 Explicit anchors and a convergent polylogarithm generator

Write $\boldsymbol{a} \vDash e$ for a composition of e. For $e \geq 1$ and $h \leq 1$, excluding $(e,h) = (1,1)$, define the finite MZV combination

$$\mathcal{Z}_{e,h} = \sum_{(a_1,\ldots,a_\ell)\vDash e}\frac{e!}{a_1!\cdots a_\ell!}\zeta(2a_1,\ldots,2a_{\ell-1},2a_\ell - h). \tag{2.15}$$

All the displayed multiple zeta values converge. Ordering the indices by their minimum, as in Hoffman's tail-product method [5], gives

$$C_{e,h} = \mathcal{Z}_{e,h} = \sum_{n\geq 1} n^h (A_{n-1}^e - A_n^e), \qquad h \leq 1. \tag{2.16}$$

The exceptional pair is divergent and is never needed. For $h = 0$, telescoping yields $\mathcal{Z}_{e,0} = \zeta(2)^e$. Formula (2.15) is classical index ordering, not a new claim about unweighted tail products.

Let I^q denote an anchored q-fold primitive:

$$(I^q f)(t) = \frac{1}{(q-1)!} \int_0^t (t-u)^{q-1} f(u)\, du, \qquad q \geq 1. \tag{2.17}$$

For $d \geq 1$ put $c_{d,j} = \binom{d}{2j} B_{2j}$ and let

$$\begin{aligned} R_d(t) = {} & \mathbf{1}_{2|d}\, \mathcal{Z}_{d,1} t - \mathbf{1}_{2\nmid d}\, \frac{d}{4} \zeta(2)^{d-1} t^2 \\ & - \sum_{j=1}^{\lfloor (d-1)/2 \rfloor} c_{d,j} \sum_{\substack{2\leq k\leq 4j \\ k\equiv d+1 \ (2)}} \mathcal{Z}_{d-2j,k-4j} \frac{t^k}{k!}. \end{aligned} \tag{2.18}$$

For $d = 1$ the first term is absent; it is not an invocation of the undefined anchor $\mathcal{Z}_{1,1}$.

Theorem 2.4 (An anchored generator at every power)**.** *The series in* (2.9) *converge for* $|t| < 2\pi$. *They are determined recursively by* $G_0 = H_0 = 0$ *and*

$$\boxed{\begin{aligned} H_d(t) = {} & R_d(t) + dI^2(\cosh(t/2) G_{d-1}(t)) \\ & - \sum_{j=1}^{\lfloor (d-1)/2 \rfloor} c_{d,j} I^{4j} H_{d-2j}(t), \qquad G_d(t) = \frac{H_d(t)}{2\sinh(t/2)}. \end{aligned}} \tag{2.19}$$

The apparent singularity at zero is removable. At each fixed d, *all moments* $M_{d,m}$ *lie in one explicitly specified finite rational span of convergent MZVs and* 1, *independent of* m. *One uniform, possibly redundant choice is the span of* 1 *and all positive-index MZVs of weight at most* $2d-1$ *with first index at least two.*

Moreover H_d *has a finite expression in polynomials in* t, *multiple zeta constants, and one-variable multiple polylogarithms at* e^{-t} *of functional depth at most* $d-1$. *This is an upper bound in a specified representation, not a minimal-depth or period-independence claim.*

Proof. Sum (2.14) with the factors $t^k/k!$. The positive lower indices give $I^{4j} H_{d-2j}$. The nonpositive lower indices give exactly the finite polynomial in (2.18), and the $k = 1$ coefficient at even d is (2.16). Thus (2.19) holds first as a formal power-series identity.

Inductively the numerator is holomorphic for $|t| < 2\pi$ and has parity $d+1$. It is $O(t)$ at even d and $O(t^2)$ at odd d. The denominator has only a simple zero at zero inside this disk. Division therefore preserves holomorphy and gives parity d. This proves convergence of the formal moment series and the asserted value of its coefficients. At fixed d only finitely many anchors from powers at most d have been introduced; their weights are at most $2d-1$. Taylor coefficients of all remaining elementary factors and integrations are rational. This proves the finite-span assertion.

For the functional representation start with real $t > 0$ and put $z = e^{-t}$. The assembled expression will then denote the analytic continuation supplied by the anchored generator on

$|t| < 2\pi$; separated terms need not be holomorphic at zero. The factor that enters the new integration step is

$$\cosh(t/2)G_{d-1}(t) = \left(\frac{1}{2} + \frac{z}{1-z}\right) H_{d-1}(t).$$

Use the convention

$$L_{a_1,\dots,a_\ell}(t) = \sum_{n_1>\cdots>n_\ell\geq 1} \frac{e^{-tn_1}}{n_1^{a_1}\cdots n_\ell^{a_\ell}}.$$

The empty word represents 1. We maintain the stronger induction invariant that each nonempty polylogarithm in H_d has first index at least two. Ordinary integrations raise that first index. For the new operation, an explicit double primitive of $t^m z L_{\boldsymbol{a}}(t)/(1-z)$ is

$$P_{m,\boldsymbol{a}}(t) = \sum_{j=0}^{m} (j+1)\frac{m!}{(m-j)!} t^{m-j} L_{j+2,\boldsymbol{a}}(t).$$

Indeed $L'_{1,\boldsymbol{a}} = -zL_{\boldsymbol{a}}/(1-z)$ and $L'_{a+1,\boldsymbol{b}} = -L_{a,\boldsymbol{b}}$. This raises depth by at most one and preserves the leading-index invariant. Its value at zero is a convergent MZV, and its first derivative is

$$P'_{m,\boldsymbol{a}}(t) = -\sum_{j=0}^{m} \frac{m!}{(m-j)!} t^{m-j} L_{j+1,\boldsymbol{a}}(t).$$

For $m > 0$ the endpoint is $-m!\zeta(m+1,\boldsymbol{a})$. For $m = 0$ the only potential divergence has exactly one leading index one.

For an admissible word $\boldsymbol{a} = (a_1,\dots,a_\ell)$ define

$$\begin{aligned} K(\boldsymbol{a}) = & -\sum_{i=1}^{\ell} \zeta(a_1,\dots,a_i,1,a_{i+1},\dots,a_\ell) \\ & -\sum_{i=1}^{\ell} \zeta(a_1,\dots,a_i+1,\dots,a_\ell), \qquad K(\varnothing) = 0. \end{aligned}$$

All these values converge, and

$$L_{1,\boldsymbol{a}}(t) = \zeta(\boldsymbol{a})L_1(t) + K(\boldsymbol{a}) + o(1).$$

To prove this, subtract $\zeta(\boldsymbol{a})L_1(t)$ termwise. The remaining series is $\sum_{n\geq 1} e^{-tn}(H_{n-1}(\boldsymbol{a}) - \zeta(\boldsymbol{a}))/n$, where $H_{n-1}(\boldsymbol{a})$ is the finite multiple harmonic sum. It is absolutely convergent at $t = 0$ because $a_1 \geq 2$. Ordering its extra index among the zeta indices gives exactly the two sums defining K. Dominated convergence proves the displayed endpoint formula.

In the complete double primitive the coefficient of the potentially divergent $L_1(t)$ is $-H_{d-1}(0) = 0$. Every remaining endpoint is therefore an explicit convergent MZV. Subtracting the value and first derivative at zero anchors the primitive. The other integrations in (2.19) do not increase depth. This proves the bound $d-1$ and fixes every integration constant without assigning values to individual divergent multiple zeta series. □

For completeness, if $F'_j = -F_{j-1}$, an unanchored q-fold primitive of $t^m F_0(t)$ is

$$(-1)^q \sum_{j=0}^{m} \binom{q+j-1}{j} \frac{m!}{(m-j)!} t^{m-j} F_{q+j}(t). \tag{2.20}$$

Subtracting its Taylor polynomial of degree $q-1$ at zero supplies the anchor, with grouped limits when necessary. Thus (2.19) is also a finite algorithm for antiderivatives and multiple-polylogarithm formulas, rather than a recurrence with unspecified constants.

2.5 A finite-cutoff implementation independent of the new recurrence

The supplied verification program does not use (2.19) to construct the comparison moments. If $F_q(N)=\sum_{n=1}^{N} n^q=\sum_a f_a N^a$, ordinary finite summation by parts instead gives

$$T_{e,q}=\sum_a f_a b_e(a)+\sum_{j=1}^{e}\binom{e}{j}\left(\sum_a f_a T_{e-j,a-2j}-T_{e-j,q-2j}\right),\qquad q\ge 0. \tag{2.21}$$

For $q<0$, products of strictly greater tail indices give

$$T_{e,q}=\sum_{\boldsymbol{a}\vDash e}\frac{e!}{a_1!\cdots a_\ell!}\zeta(2a_1,\dots,2a_\ell,-q),\qquad e\ge 1.$$

The remaining base cases are $T_{0,q}=0$ for $q\ge 0$, $T_{0,-1}=\gamma$, and $T_{0,-p}=\zeta(p)$ for $p\ge 2$. The power e decreases at every recursive step. This proves existence of the constant coefficients used above and supplies a terminating independent exact verification route.

2.6 The same cancellation for every even Hurwitz order

Fix $q\ge 1$ and put $p=2q$. In this subsection the symbols M,C,G,H refer to the fixed tail $T_n=\zeta(p,n+1)=\psi^{(p-1)}(n+1)/(p-1)!$, with $x_n=n+1/2$. Thus $T_{n-1}=T_n+n^{-p}$. Define

$$\eta_0=\frac{1}{p-1},\qquad \eta_j=\frac{(p)_{2j-1}}{(2j)!}B_{2j}(1/2)\quad(j\ge 1),\qquad h_{d,j}=\sum_{j_1+\cdots+j_d=j}\prod_{\nu=1}^{d}\eta_{j_\nu}.$$

The centered Hurwitz asymptotics give $T_n^d\sim\sum_{j\ge 0}h_{d,j}x_n^{-(p-1)d-2j}$. For $m\ge 0$, $m\equiv d\pmod 2$, the regular continued moment is the ordinary absolutely convergent sum

$$M_{d,m}=\sum_{n\ge 0}\left\{x_n^m T_n^d-\sum_{0\le j\le (m-(p-1)d)/2}h_{d,j}x_n^{m-(p-1)d-2j}\right\}. \tag{2.22}$$

Indeed the next term is $O(x_n^{-2})$, while every removed term has continued sum $\zeta(-2r,1/2)=0$. The meromorphic family $\sum T_n^d x_n^{-s}$ has its poles at $1-(p-1)d-2j$; the same-sign convolution of the η_j shows that their residues $h_{d,j}$ are nonzero.

For every integer k put

$$C_{d,k}=\mathrm{CT}_N\left\{\sum_{n=1}^{N}n^k(T_{n-1}^d-T_n^d)+N^kT_N^d\right\},\qquad G_d(t)=\sum_{m\equiv d\ (2)}M_{d,m}\frac{t^m}{m!}.$$

Subtracting finitely many asymptotic terms proves existence of each cutoff constant. The latter sum has $m\ge 0$. Set $H_d=2\sinh(t/2)G_d$ and $G_0=H_0=0$. The centered Faulhaber identity used above gives

$$H_d(t)=\sum_{\substack{k\ge 1\\ k\equiv d+1\ (2)}}C_{d,k}\frac{t^k}{k!}. \tag{2.23}$$

For $e\ge 1$ and $h<p$, except $(e,h)=(1,p-1)$, the needed anchors are

$$\mathcal{Z}_{e,h}^{(q)}=\sum_{(a_1,\dots,a_\ell)\vDash e}\frac{e!}{a_1!\cdots a_\ell!}\zeta(pa_1,\dots,pa_{\ell-1},pa_\ell-h)=C_{e,h}. \tag{2.24}$$

Here $\vDash$ denotes a composition. Ordering the e tail indices by their minimum proves this formula: the summand is $n^h(T_{n-1}^e-T_n^e)=O(n^{h-(p-1)e-1})$ and the boundary term tends to zero. The excluded pair never has the required parity.

Write $c_{d,j} = \binom{d}{2j} B_{2j}$ and define the finite polynomial

$$R_d(t) = \sum_{\substack{1\le k<p \\ k\equiv d+1 \ (2)}} \mathcal{Z}^{(q)}_{d,k} \frac{t^k}{k!} - \mathbf{1}_{2\mid d} \frac{d}{2} \zeta(p)^{d-1} \frac{t^p}{p!} - \sum_{j=1}^{\lfloor (d-1)/2 \rfloor} c_{d,j} \sum_{\substack{p\le k\le 2pj \\ k\equiv d+1 \ (2)}} \mathcal{Z}^{(q)}_{d-2j,k-2pj} \frac{t^k}{k!}. \tag{2.25}$$

Theorem 2.5 (Every power of every even-order tail)**.** *For $k \ge p$ with $k \equiv d+1 \pmod 2$,*

$$C_{d,k} = d(k-p)![t^{k-p}](\cosh(t/2)G_{d-1}(t)) - \frac{d}{2}\delta_{k,p}\zeta(p)^{d-1} - \sum_{j=1}^{\lfloor (d-1)/2 \rfloor} c_{d,j} C_{d-2j,k-2pj}. \tag{2.26}$$

With the anchored primitives of (2.17), its generator is

$$H_d = R_d + dI^p(\cosh(t/2)G_{d-1}) - \sum_{j=1}^{\lfloor (d-1)/2 \rfloor} c_{d,j} I^{2pj} H_{d-2j}, \qquad G_d = \frac{H_d}{2\sinh(t/2)}. \tag{2.27}$$

Both functions are holomorphic for $|t| < 2\pi$. At fixed (q,d), all the moments belong to a fixed finite rational span of 1 and convergent MZVs of weights at most $2qd-1$. The generator H_d admits a finite multiple-polylogarithm representation of functional depth at most $d-1$, with all constants fixed by the anchored primitives.

Proof. The uncentered formal asymptotic series is

$$\mathcal{A}_q(u) = u^{p-1}E_q(u) - \tfrac{1}{2}u^p, \qquad E_q(u) = \frac{1}{p-1} + \sum_{j\ge 1} \frac{(p)_{2j-1}B_{2j}}{(2j)!}u^{2j}.$$

Its boundary combination satisfies the exact formal identity

$$\mathcal{A}_q^d + \frac{d}{2}u^p\mathcal{A}_q^{d-1} + \sum_{j=1}^{\lfloor d/2 \rfloor} c_{d,j}u^{2pj}\mathcal{A}_q^{d-2j} = u^{pd}B_d\left(\frac{1}{2} + \frac{E_q(u)}{u}\right). \tag{2.28}$$

Since E_q is even, Bernoulli reflection makes the right side have parity d. Thus the required opposite-parity coefficients vanish. Apply the finite polynomial identity (2.11) with $a = T_n$, $h = n^{-p}$ before taking constant coefficients. For $r = k-p \ge 0$, the symmetric term has the exact cutoff shift

$$\sum_{n=1}^{N} n^r(T_{n-1}^{d-1} + T_n^{d-1}) = \sum_{n=0}^{N-1}((n+1)^r + n^r)T_n^{d-1} + N^r T_N^{d-1} - \delta_{r,0}\zeta(p)^{d-1}.$$

The centered bracket has regular parity and gives $2r![t^r](\cosh(t/2)G_{d-1})$. Equation (2.28) cancels the remaining boundaries, proving (2.26). For $d = 1$ the polynomial sums have zero constant coefficient, in agreement with $G_0 = 0$.

Summing the recurrence gives (2.27): the low positive indices give the first term of R_d, and the nonpositive lower indices give its last term. Induction gives holomorphy on the stated disk; H_d has parity $d+1$ and vanishes at zero, so the division is removable. Each low anchor has weight $pe - h \le pd - 1$; each last-term anchor has weight $p(d-2j) - (k-2pj) = pd - k \le pd - p$. All other Taylor operations are rational. Finally $\cosh(t/2)G_{d-1} = (1/2 + z/(1-z))H_{d-1}$, $z = e^{-t}$; the same primitive construction as in Theorem 2.4 raises functional depth by at most one. Increasing the number of anchored integrations from 2 to p adds no depth. □

This extends the inspected two-tail and trigamma-power results to every repeated even order. It does not yet address unequal orders among three or more factors. No priority claim beyond the inspected source corpus is intended.

Corollary 2.6 (An ordinary cubic identity at every even order)**.** *For every $q \geq 1$, with $T_n = \zeta(2q, n+1)$,*

$$\boxed{\sum_{n\geq 0}\Big\{((n+1)^{2q} - n^{2q})T_n^3 - 3T_n^2\Big\} = -\frac{3\zeta(2q)^2 + \zeta(4q)}{2}.} \tag{2.29}$$

In particular, for $x_n = n + 1/2$ and $T_n = \psi^{(3)}(n+1)/6$,

$$\boxed{\sum_{n\geq 0}((4x_n^3 + x_n)T_n^3 - 3T_n^2) = -\frac{9}{4}\zeta(8).} \tag{2.30}$$

Both identities are absolutely convergent without subtraction.

Proof. Put $p = 2q$. The cubic case of (2.11), followed by finite summation by parts, gives the exact cutoff identity

$$\sum_{n=0}^{N-1}\Big\{((n+1)^p - n^p)T_n^3 - 3T_n^2\Big\} = N^p T_N^3 + \frac{3}{2}T_N^2 - \frac{3}{2}\zeta(p)^2 - \frac{1}{2}H_N^{(2p)}.$$

Here the cubic difference is $n^p(T_{n-1}^3 - T_n^3) = \frac{3}{2}(T_{n-1}^2 + T_n^2) - \frac{1}{2}n^{-2p}$. Since $T_n = O((n+1)^{1-p})$, the two boundary terms tend to zero and each series on the left converges absolutely for $p \geq 2$. Letting $N \to \infty$ proves (2.29). For $p = 4$, use $(n+1)^4 - n^4 = 4x_n^3 + x_n$ and $\zeta(4)^2 = 7\zeta(8)/6$ to obtain (2.30). □

Corollary 2.7 (Complex orders and a critical cubic value)**.** *Put $T_n(p) = \zeta(p, n+1)$, using the real logarithm of positive integers in all powers and setting $0^p = 0$ in the domain below. For every complex p with $\Re p > 3/2$,*

$$\sum_{n\geq 0}\Big\{((n+1)^p - n^p)T_n(p)^3 - 3T_n(p)^2\Big\} = -\frac{3\zeta(p)^2 + \zeta(2p)}{2}. \tag{2.31}$$

At the real critical order $p = 3/2$, the combined series also converges absolutely, but its value has an additional boundary term:

$$\boxed{\sum_{n\geq 0}\Big\{((n+1)^{3/2} - n^{3/2})T_n(3/2)^3 - 3T_n(3/2)^2\Big\} = 8 - \frac{3\zeta(3/2)^2 + \zeta(3)}{2}.} \tag{2.32}$$

Proof. The preceding cutoff identity remains exact for $\Re p > 1$. For fixed p in this half-plane, $T_N(p) = N^{1-p}/(p-1) + O(N^{-\Re p})$. If $\Re p > 3/2$, each series on its left converges absolutely and $N^p T_N(p)^3 = O(N^{3-2\Re p}) \to 0$, proving (2.31).

At $p = 3/2$, put $x = n + 1/2$. The centered expansions are

$$T_n(3/2) = 2x^{-1/2} - \frac{1}{16}x^{-5/2} + O(x^{-9/2}), \qquad (n+1)^{3/2} - n^{3/2} = \frac{3}{2}x^{1/2} - \frac{1}{64}x^{-3/2} + O(x^{-7/2}).$$

Consequently the combined summand in (2.32) is $-\frac{1}{2}x^{-3} + O(x^{-5})$, proving absolute convergence. In the exact cutoff formula, $N^{3/2}T_N(3/2)^3 \to 8$, $T_N(3/2)^2 \to 0$, and $H_N^{(3)} \to \zeta(3)$. These limits give the stated critical value. □

3 Fifth and sixth powers: finite coordinate spaces and explicit identities

We now make the all-power theorem concrete. Retain the classical constants

$$Q_5 = 4\zeta(2)\zeta(3) - 10\zeta(5), \tag{3.1}$$

$$Q_7 = 51\zeta(3)\zeta(4) + 10\zeta(2)\zeta(5) - \frac{301}{4}\zeta(7). \tag{3.2}$$

The latter is $\sum_{n\geq 0} A_n^4$, by Hoffman's quartic evaluation with the $n = 0$ term included [5, 2].

3.1 The lower-power inputs

The universal recurrence starts with

$$H_1(t) = -\frac{t^2}{4}, \tag{3.3}$$

$$H_2(t) = 3\zeta(3)t + 2I^2(\cosh(t/2)G_1(t)), \tag{3.4}$$

$$H_3(t) = 3I^2(\cosh(t/2)G_2(t)) - \frac{17}{8}\zeta(4)t^2 - \frac{\zeta(2)}{48}t^4 + \frac{t^6}{2880}, \tag{3.5}$$

$$H_4(t) = Q_7 t + \frac{Q_5 t^3}{6} + 4I^2(\cosh(t/2)G_3(t)) - I^4 H_2(t). \tag{3.6}$$

These reproduce the existing quadratic, cubic, and quartic generators. For example $\mathcal{Z}_{2,-1} = 2\zeta(2,3) + \zeta(5) = -Q_5$ and $\mathcal{Z}_{e,0} = \zeta(2)^e$ account for their exceptional terms.

3.2 The quintic generator and its eight coordinates

Define the one new weight-eight combination

$$K_8 = \frac{5}{6}\mathcal{Z}_{3,-2} + \frac{289}{32}\zeta(8), \quad \mathcal{Z}_{3,-2} = 6\zeta(2,2,4) + 3\zeta(2,6) + 3\zeta(4,4) + \zeta(8). \tag{3.7}$$

This is a specified convergent MZV coordinate, not a claim of a new transcendental constant or of its irreducibility.

Theorem 3.1 (Every regular quintic moment)**.** *For $|t| < 2\pi$,*

$$\boxed{\begin{aligned} H_5(t) = 5I^2(\cosh(t/2)G_4(t)) - \frac{5}{3}I^4 H_3(t) - K_8 t^2 - \frac{19}{64}\zeta(6)t^4 \\ + \frac{\zeta(4)}{4320}t^6 + \frac{\zeta(2)}{241920}t^8 - \frac{t^{10}}{43545600}. \end{aligned}} \tag{3.8}$$

All the moments $M_{5,2r+1}$ lie in

$$\operatorname{span}_{\mathbb{Q}}\{K_8, \zeta(6), Q_7, Q_5, \zeta(2), \zeta(3), \zeta(4), 1\}. \tag{3.9}$$

The coefficient matrix of the first eight moments in this ordered list has determinant

$$\frac{672072587373661}{748313054183581286400} \neq 0. \tag{3.10}$$

Thus eight is the exact rank of these specified rational coefficient rows. No arithmetic independence of the eight evaluated constants is asserted.

Proof. At $d = 5$, the two Bernoulli coefficients in (2.19) are $5/3$ and $-1/6$. Insert the anchors $\mathcal{Z}_{3,-2}$, $\mathcal{Z}_{3,0} = \zeta(2)^3$, and $\mathcal{Z}_{1,-2j} = \zeta(2j+2)$. Euler's even-zeta evaluations give $\zeta(2)^3 = 35\zeta(6)/8$ and $\zeta(2)^4 = 175\zeta(8)/24$. Finally insert $I^8 H_1 = -t^{10}/7257600$. Collecting these finite terms gives (3.8). The span follows from the lower-power generators. Exact rational coefficient extraction gives (3.10); the complete matrix and an executable determinant calculation accompany the article. □

The first three formulas, useful independently of the generator, are

$$M_{5,1} = -K_8 + \frac{5}{2}Q_7, \tag{3.11}$$

$$M_{5,3} = \frac{1}{4}K_8 + \frac{5}{12}Q_5 - \frac{5}{12}Q_7 + \frac{15}{2}\zeta(3) - \frac{85}{24}\zeta(4) - \frac{57}{32}\zeta(6), \tag{3.12}$$

$$\begin{aligned} M_{5,5} = &-\frac{7}{48}K_8 - \frac{5}{72}Q_5 + \frac{47}{288}Q_7 - \frac{1}{12}\zeta(2) - \frac{5}{2}\zeta(3) + \frac{157}{144}\zeta(4) \\ &+ \frac{95}{64}\zeta(6) - \frac{1}{12}. \end{aligned} \tag{3.13}$$

The first two are unsubtracted sums. The third is $\sum_{n\geq 0}(x_n^5 A_n^5 - 1)$.

Corollary 3.2 (Two ordinary quintic evaluations)**.** *With $x = x_n$ and $A = A_n$, both series below converge absolutely:*

$$\begin{aligned} &\sum_{n\geq 0}\big((23x + 232x^3 + 240x^5)A^5 - 240\big) \\ &\qquad = 80Q_5 - 20\zeta(2) + 1140\zeta(3) - 560\zeta(4) - 57\zeta(6) - 20, \end{aligned} \tag{3.14}$$

$$\boxed{\begin{aligned} &\sum_{n\geq 0}\Big[(195x + 4056x^3 + 11568x^5 + 5376x^7)A^5 - 5376x^2 - 9328\Big] \\ &\qquad = -748\zeta(2) + 7884\zeta(3) - 5380\zeta(4) + 171\zeta(6) - 748. \end{aligned}} \tag{3.15}$$

Proof. The first identity is $23M_{5,1} + 232M_{5,3} + 240M_{5,5}$. Its K_8 and Q_7 coefficients cancel. The second is $195M_{5,1} + 4056M_{5,3} + 11568M_{5,5} + 5376M_{5,7}$. The three coefficients of K_8, Q_7, Q_5 cancel. Its subtraction polynomial is $5376(x^2 - 5/12) + 11568 = 5376x^2 + 9328$.

There is a particularly small independent MZV check of the second formula. The finite-cutoff recursion (2.21) reduces its left side to

$$\begin{aligned} &-748\zeta(2) + 5256\zeta(2,1) - 3840\zeta(2,2) + 240\zeta(2,2,2) \\ &\quad + 120\big(\zeta(2,4) + \zeta(4,2)\big) + 2628\zeta(3) - 2500\zeta(4) + 36\zeta(6) - 748. \end{aligned}$$

Use $\zeta(2,1) = \zeta(3)$, $\zeta(2,2) = 3\zeta(4)/4$, $\zeta(2,2,2) = 3\zeta(6)/16$, and $\zeta(2,4)+\zeta(4,2) = 3\zeta(6)/4$. These are Euler, all-twos, and elementary stuffle identities. The displayed result follows. Absolute convergence follows directly from (2.3). □

The values of (3.14) and (3.15) begin, respectively,

$$456.551810834060785171538560838 6\ldots, \qquad 1849.672608741185304145251799308\ldots.$$

These decimals are diagnostic illustrations, not inputs to the proof.

3.3 The sextic generator and eleven coordinates

Define

$$R_{11} = \mathcal{Z}_{6,1}, \qquad K_9 = \mathcal{Z}_{2,-5} - 5\mathcal{Z}_{4,-1}, \qquad J_7 = \mathcal{Z}_{2,-3} = 2\zeta(2,5) + \zeta(7). \tag{3.16}$$

Formula (2.15) specifies every term of these finite convergent MZV combinations.

Theorem 3.3 (Every regular sextic moment)**.** *For $|t| < 2\pi$,*

$$\boxed{\begin{aligned} H_6(t) = R_{11}t + 6I^2(\cosh(t/2)G_5(t)) - \frac{5}{2}I^4H_4(t) + \frac{1}{2}I^8H_2(t) \\ + \frac{K_9}{12}t^3 + \frac{J_7}{240}t^5 - \frac{Q_5}{10080}t^7. \end{aligned}} \tag{3.17}$$

All $M_{6,2r}$ belong to the rational span of the ordered list

$$(R_{11}, K_9, J_7, K_8, \zeta(6), Q_7, Q_5, \zeta(2), \zeta(3), \zeta(4), 1). \tag{3.18}$$

The first eleven coefficient rows have determinant

$$-\frac{55732534349435909753872230 7}{13592747873561186778319872000000 0} \neq 0. \tag{3.19}$$

This is again a formal coefficient-rank statement.

Proof. The nonzero coefficients at $d = 6$ in (2.19) are $5/2$ and $-1/2$. The $k = 1$ anchor is R_{11}; the remaining negative-index anchors combine to K_9, J_7, and $\mathcal{Z}_{2,-1} = -Q_5$. This gives (3.17) directly. The span and determinant follow by rational coefficient extraction. □

For example,

$$\begin{aligned} M_{6,0} &= R_{11}, \\ M_{6,2} &= -2K_8 + \frac{1}{6}K_9 + 5Q_7 - \frac{1}{12}R_{11}, \\ M_{6,4} &= \frac{1}{10}J_7 + \frac{2}{5}K_8 - \frac{1}{12}K_9 + \frac{1}{2}Q_5 - \frac{5}{4}Q_7 + \frac{7}{240}R_{11} \\ &\quad + 9\zeta(3) - \frac{17}{4}\zeta(4) - \frac{171}{80}\zeta(6). \end{aligned} \tag{3.20}$$

One further elimination gives an ordinary sixth-power identity:

$$\begin{aligned} &\sum_{n\geq 0}\Big[(35 + 3360x^2 + 29792x^4 + 50176x^6 + 16128x^8)A^6 - 16128x^2 - 42112\Big] \\ &\quad = 9184Q_5 - 64Q_7 - 2928\zeta(2) + 111216\zeta(3) - 55368\zeta(4) - 9576\zeta(6) - 2928. \end{aligned} \tag{3.21}$$

It is seven times the combination with coefficients $(5, 480, 4256, 7168, 2304)$ of $M_{6,0}, \ldots, M_{6,8}$. Its value is $33200.31747342681779918605867754\ldots$.

4 The resonant parity and all Laurent coefficients

The regular moments do not determine spectral derivatives at the interlaced poles. The same explicit subtractions nevertheless give the complete Laurent family, including all Stieltjes terms. This addresses the normalization part of the quartic report's Research question R5 and applies to every power.

Use the convention

$$\zeta(1+\varepsilon,a)=\frac{1}{\varepsilon}+\sum_{q\ge 0}\frac{(-1)^q\gamma_q(a)}{q!}\varepsilon^q. \tag{4.1}$$

For $d\ge 1$, $r\ge 0$, put $s_0=1-d-2r$ and

$$F_d(s_0+\varepsilon)=\frac{h_{d,r}}{\varepsilon}+\sum_{q\ge 0}\Lambda_{d,r,q}\frac{\varepsilon^q}{q!}.$$

Theorem 4.1 (Exact logarithmic sums at every resonance)**.** *For every $q\ge 0$,*

$$\boxed{\begin{aligned}\Lambda_{d,r,q}=(-1)^q\sum_{n\ge 0}(\log x_n)^q\left[x_n^{d+2r-1}A_n^d-\sum_{j=0}^{r}h_{d,j}x_n^{2r-1-2j}\right]\\ +\sum_{j=0}^{r-1}h_{d,j}\zeta^{(q)}(1+2j-2r,1/2)+(-1)^q h_{d,r}\gamma_q(1/2).\end{aligned}} \tag{4.2}$$

The series is ordinary and absolutely convergent. The Taylor series of $F_d(s_0+\varepsilon)-h_{d,r}/\varepsilon$ has exact radius 2.

Proof. The square bracket is $O(x_n^{-3})$. On a compact subset K of $\Re\varepsilon>-2$, its product with $x_n^{-\varepsilon}$ has qth derivative bounded by

$$O_{K,q}\Big(x_n^{-3-\min_{\varepsilon\in K}\Re\varepsilon}(1+|\log x_n|)^q\Big).$$

This bound is summable, so the remainder and all fixed derivatives converge normally there. Add

$$\sum_{j=0}^{r-1}h_{d,j}\zeta(1+2j-2r+\varepsilon,1/2)+h_{d,r}(\zeta(1+\varepsilon,1/2)-1/\varepsilon).$$

This is the required pole-subtracted meromorphic family on a common half-plane, hence by continuation near zero. Differentiating there gives (4.2), including the sign and normalization of the Stieltjes term. There are no poles in $|\varepsilon|<2$, while $\varepsilon=-2$ is an actual pole with nonzero residue $h_{d,r+1}$. This proves the radius assertion. □

At the first resonance the formula is especially short:

$$\Lambda_{d,0,q}=(-1)^q\left[\sum_{n\ge 0}(\log x_n)^q\left(x_n^{d-1}A_n^d-x_n^{-1}\right)+\gamma_q(1/2)\right]. \tag{4.3}$$

All half-shift Stieltjes constants can be converted to the ordinary constants $\gamma_q=\gamma_q(1)$ by

$$\gamma_q(1/2)=\frac{2(-1)^q(\log 2)^{q+1}}{q+1}+2\sum_{j=0}^{q}\binom{q}{j}(-\log 2)^{q-j}\gamma_j-\gamma_q. \tag{4.4}$$

This follows by expanding $\zeta(s,1/2)=(2^s-1)\zeta(s)$ at $s=1$.

The theorem gives a locally analytic, explicitly anchored generator in the spectral displacement ε. It does not identify a finite arithmetic span as r or the logarithmic order q varies; that is a further reduction problem. In particular one must not differentiate the discrete coefficient formulas for $M_{d,m}$ as though they were identities at arbitrary complex m.

5 Completion of the smaller-order-ten reconstruction theorem

Two incoming reports leave complementary parts of the same problem open. The *Alternating Tails and Spectral Defects* report [3] proves the smaller-order-ten statement outside one residue class modulo 2048; the *Quartic Spectral Reconstruction* report [2] proves it for all sufficiently large larger orders. Their thresholds leave exactly one matrix unaccounted for. We prove that this matrix is nonsingular by a short exact certificate and hence complete the entire smaller-order-ten family. The finite polynomial identities and the sufficient threshold used from those reports are replayed independently below. Their underlying reductions remain attributed to those reports.

This completion concerns formal rational coordinate rank. It implies identities among actual polygamma sums, but does not assert arithmetic independence of evaluated zeta values. The unrestricted unequal-order conjecture remains open.

5.1 Moment coordinates and the precise theorem

For integers $s \geq 2$ put

$$A_s(n) = \zeta(s, n+1) = \zeta(s) - H_n^{(s)}, \qquad x_n = n + \tfrac{1}{2}.$$

For an integer $b \geq 6$, initially define

$$D_{10,2b}(w) = \sum_{n \geq 0} A_{10}(n) A_{2b}(n) x_n^{-w}, \qquad \operatorname{Re} w > -2b - 7.$$

The centered Hurwitz expansion contains only powers $x_n^{-2b-8-2j}$, so continuation by subtraction is regular at every nonpositive even integer. Write $D_r = D_{10,2b}(-2r)$. Define the mutually inverse finite transforms

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r, \tag{5.1}$$

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j. \tag{5.2}$$

Finite summation by parts gives, for every $j \geq b$,

$$C_j = \sum_{k=0}^{b-1} (M_{5,b}^{\infty})_{j-b,k} Z_k, \qquad Z_k = \zeta(2k), \quad Z_0 = -\tfrac{1}{2},$$

where, for $t \geq 0$ and $0 \leq k < b$,

$$(M_{5,b}^{\infty})_{tk} = \mu_{t+k-4} \left\{ \frac{(2b+2t-9)!}{(2b-2k)!} + \mathbf{1}_{k<5} \frac{(2t+1)!}{(10-2k)!} \right\}, \qquad \mu_j = \begin{cases} B_{2j}/(2j)!, & j \geq 0, \\ 0, & j < 0. \end{cases} \tag{5.3}$$

Let $M_{5,b}$ be its first b rows. Equivalently, each entry is

$$B_{2(t+k-4)} \left\{ \frac{\binom{2t+2}{10-2k}}{2t+2} + \frac{\binom{2b+2t-8}{2b-2k}}{2b+2t-8} \right\},$$

where a negative Bernoulli index gives zero, and a binomial summand is omitted unless its lower index lies between two and its upper index. This last convention is essential.

The formal coordinate space here has independent labels $C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}$. Its evaluation map assigns the indicated complex numbers to those labels. In these coordinates the first $2b$ transformed moments have matrix $\operatorname{diag}(I_b, M_{5,b})$. Since the diagonal of (5.2) is $1/(2r+1)$, the first $2b$ original moments have determinant $\det M_{5,b}/(4b-1)!!$.

Theorem 5.1 (Every larger even order at smaller order ten)**.** *For every integer $b \geq 6$, $M_{5,b}$ is nonsingular. Consequently*

$$D_{10,2b}(0), D_{10,2b}(-2), \ldots, D_{10,2b}(-2(2b-1))$$

have full formal coordinate rank $2b$, and every subsequent centered even moment is an explicit rational linear combination of this prefix.

5.2 The five-dimensional contraction and its normalization

For $0 \leq i, \ell \leq 4$ set

$$n_i = b - 5 + i, \qquad N_{i\ell} = 2(n_i - \ell) + 1, \qquad R_\ell = 2b - 8 + 2\ell,$$

$$F_{i\ell} = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n_i - r)(2n_i + 3 + r).$$

An empty product is one. Define finite integer-polynomial coefficients d_{ih} by the Newton expansion

$$(2n_i + 2 - y)(4n_i + 2 - y)^{\underline{2i}} = \sum_{h=0}^{2i+1} d_{ih} y^{\underline{h}}. \tag{5.4}$$

Thus explicitly

$$d_{ih} = \frac{1}{h!} \sum_{j=0}^{h} (-1)^{h-j} \binom{h}{j} (2n_i + 2 - j)(4n_i + 2 - j)^{\underline{2i}}.$$

For independent variables $U_1, \ldots, U_9$, let

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)!N_{i\ell}!} \sum_{k=0}^{i} d_{i,2k+1} N_{i\ell}^{\underline{2k+1}} U_{5-i+\ell+k}. \tag{5.5}$$

Ratios of factorials have fixed index difference and mean the corresponding finite rational products in b.

For $b \geq 10$, the exact reduction is

$$\rho_{5,b} := \frac{\det M_{5,b}}{\det R \det E \det H_{5,b}} = \frac{\det \mathcal{A}(b; 2B_{2b-2}, \ldots, 2B_{2b-18})}{\det F(b)}, \tag{5.6}$$

where

$$R_{tt} = (2b + 2t - 9)!, \qquad E_{kk} = \frac{1}{(2b-2k)!}, \qquad (H_{5,b})_{tk} = \mu_{t+k-4}.$$

All three determinants in the denominator are nonzero.

Here are details sufficient to check the contraction without an assumed inverse-Hankel formula. The monic polynomial rows

$$\pi_n(x) = \sum_{t=0}^{n} \frac{2(2n+2t+1)!(2n+1)!}{(4n+2)!(2n-2t+1)!(2t)!} x^t$$

are orthogonal for the functional $\mathcal{N}(x^r) = \mu_{r+1}$:

$$\mathcal{N}(x^j \pi_n) = 0 \quad (0 \leq j < n), \qquad \mathcal{N}(x^n \pi_n) = h_n, \quad h_n = \frac{((2n+1)!)^2}{(4n+2)!(4n+3)!} > 0.$$

These follow by coefficient comparison in the elementary Padé identity

$$e^t P_N(-t) - P_N(t) = \frac{(-1)^N (N!)^2}{(2N)!(2N+1)!} t^{2N+1} + O(t^{2N+2}),$$

$$P_N(t) = \sum_{k=0}^{N} \frac{(2N-k)!N!}{(2N)!k!(N-k)!} t^k, \qquad N = 2n+1.$$

The coefficient cancellation through degree $2N$ is the finite binomial identity $\sum_k (-1)^k \binom{N}{k}(1+x)^{2N-k} = x^N(1+x)^N$; the next coefficient is an elementary beta integral.

Replacing every row of $H_{5,b} + U^{\mathrm{pert}}$ by these monic rows is a unit lower triangular operation. Its final five rows vanish in columns $k \geq 5$. The unperturbed retained entries, after column reversal and division by $c_n = (2n+1)(2n+2)!/(4n+2)!$, are precisely $F_{i\ell}$. For the perturbation apply

$$\sum_{j=0}^{N} \binom{N}{j} B_j (N-j)^{\underline{h}} = N^{\underline{h}} B_{N-h}(1)$$

to (5.4). Because $b \geq 10$, every surviving index $N-h$ is at least two. The even h give vanishing odd Bernoulli numbers; restoring the omitted term $j=1$, with $B_1 = -1/2$, contributes a second $F_{i\ell}$. This proves (5.5), including both factors of two. Carrying the same row operation out on $H_{5,b}$ alone gives (5.6).

For completeness, $H_{5,b}$ is nonsingular: after conjugating by $\mathrm{diag}(1,-1,1,\dots)$ and splitting off its first three indices, it has the form

$$\begin{pmatrix} ee^{\mathsf{T}} & B \\ B^{\mathsf{T}} & -P \end{pmatrix}, \qquad e = (0,0,1)^{\mathsf{T}},$$

where P is positive definite. Its entries are moments on the infinite positive support $\{(2\pi m)^{-2} : m \geq 1\}$. The first two rows of B have a nonsingular antitriangular two-by-two minor. Therefore $ee^{\mathsf{T}} + BP^{-1}B^{\mathsf{T}}$ is positive definite, which proves nonsingularity by the Schur complement. This argument also shows that the unperturbed contraction F is nonsingular.

5.3 The inherited congruence, with its finite certificate replayed

Define

$$\begin{aligned} D_{10}(b) = {} & 4096b^{12} - 110592b^{11} + 1236992b^{10} - 7303680b^9 \\ & + 23705088b^8 - 38140416b^7 + 14925056b^6 + 24490080b^5 \\ & - 4469984b^4 - 15838992b^3 - 2571948b^2 + 4244400b + 1488375. \end{aligned}$$

The exact five-by-five polynomial identities are

$$\begin{aligned} \det F = {} & \frac{16384}{4465125}(b-3)(b-2)^2(b-1)(2b-7)(2b-5)^2(2b-3) \\ & \cdot (4b-15)(4b-13)(4b-11)^2(4b-9)^2(4b-7)^2(4b-5)(4b-3)D_{10}(b), \end{aligned} \tag{5.7}$$

$$\frac{\det \mathcal{A}}{\det F} = \frac{P_{10}(b; U_1, \dots, U_9)}{138240(2b-7)(2b-5)D_{10}(b)}, \tag{5.8}$$

where P_{10} is an integer polynomial. Equations (5.5) and (5.8) define it exactly; its supplied sparse certificate contains 10,403 monomials, grouped into 439 monomials in the U variables. Their coefficients have degree at most 24 in b.

The replay forms the row denominators of (5.5), constructs the 120 signed permutation products of the cleared matrix, and cross-multiplies (5.8) in $\mathbb{Q}[b, U_1, \dots, U_9]$. Equality is checked coefficient by coefficient. Thus polynomial integrality and the displayed determinant identity are finite rational certificates, without interpolation.

Put

$$\mathcal{B}(x) = 1 + 20068x - 26384x^2 + 2816x^3 - 19712x^4 + 4096x^5 - 24576x^6.$$

The uniform Bernoulli congruence is

$$2B_{2m} \equiv \mathcal{B}(m) \pmod{65536} \quad (m \geq 12). \tag{5.9}$$

All congruences are in $\mathbb{Z}_{(2)}$, the rationals with odd denominator. One direct proof starts with Faulhaber's formula at 256. The terms other than $256B_{2m}$ have valuation at least 23; after division by 128, the even summands vanish at the required precision because $2m \geq 24$. For odd u, $8 \mid u^2 - 1$, so expansion of $u^{2m} = (1 + (u^2 - 1))^m$ gives

$$2B_{2m} \equiv \frac{1}{128} \sum_{u=1,3,\dots,255} \sum_{j=0}^{7} \binom{m}{j} (u^2 - 1)^j \pmod{65536}.$$

The omitted terms have valuation at least $3 \cdot 8 - 7 = 17$. Expanding this finite average in $\mathbb{Q}[m]$ and reducing every coefficient modulo 65536 gives (5.9). The replay checks this coefficientwise, including the denominators of $\binom{m}{j}$. The bound $v_2(B_{2j}) \geq -1$ used in the Faulhaber step follows inductively from the odd-denominator recurrence (5.14) below.

The finite polynomial substitution then gives

$$P_{10}(b; \mathcal{B}(b-1), \dots, \mathcal{B}(b-9)) \equiv 32\mathcal{R}(b) \pmod{65536}, \tag{5.10}$$

where

$$\mathcal{R}(b) = -512b^8 + 128b^7 - 928b^6 - 816b^5 + 464b^4 + 352b^3 + 118b^2 + 217b + 694.$$

The replay verifies every coefficient of (5.10) using modular polynomial multiplication. The linear coefficient of $\mathcal{R}$ is odd and every coefficient of degree at least two is even. Consequently

$$v_2(\mathcal{R}(x) - \mathcal{R}(y)) = v_2(x - y) \quad (x \neq y).$$

Indeed the quotient by $x - y$ is odd. Direct substitution gives $\mathcal{R}(962) \equiv 0 \pmod{2048}$, so 962 is its unique root modulo 2048.

Since $D_{10}(b)$ is odd at every integer b, the denominator in (5.8) has valuation 10. For $b \geq 21$ all nine Bernoulli indices are in the range of (5.9). If $b \not\equiv 962 \pmod{2048}$, equations (5.9)–(5.10) therefore prove

$$v_2(\rho_{5,b}) = v_2(b - 962) - 5. \tag{5.11}$$

For $6 \leq b \leq 20$, direct rational determinants of the original matrix (5.3) give the respective valuations

$$-4, -4, -6, -5, -2, -5, -4, -5, -3, -5, -4, -5, -1, -5, -4.$$

The direct matrices, their determinants, and their normalizations are included in the replay record. In particular, the contraction is never used below its valid threshold $b \geq 10$.

5.4 A transparent sufficient large-order threshold

We specialize the incoming eventual-rank proof to $a = 5$ and replace its large rational constant by an elementary upper bound. This makes the threshold independently reviewable.

Define

$$J(b) = \prod_{i=0}^{4}(2n_i+1)(2n_i+2), \qquad E(b) = \prod_{i,\ell=0}^{4}\prod_{r=1}^{\max(0,2i-4\ell-1)}(R_\ell + r), \quad D(b) = 10!J(b)E(b).$$

Here $\deg J = 10$, $\deg E = 20$, and $D\mathcal{A}$ has integer polynomial entries. Put $\mathscr{P} = \det(D\mathcal{A})$ and let $\mathscr{P}_{[5]}$ be its homogeneous part of degree five in the U variables. Every entry has degree at most 46 in b; hence $\deg_b \mathscr{P} \le 230$. The largest coefficient norm among the 25 entries, with all coefficients counted in $\mathbb{Z}[b, U_1, \dots, U_9]$, is

$$E^* = 602847821074996042728613866361551228764160 < 10^{42}.$$

The separate verifier expands the 25 entries, records all 25 norms and degrees, and reproduces this integer. The full determinant coefficient norm is at most $K = 5!(E^*)^5 < 10^{213}$.

Let $F_0 = (2b-18)!$ and $V_j = U_jF_0/(2b-2j)!$. From (5.4), $d_{i1} = -(i+1)(4n_i+2)^{\underline{2i}}$. Cancelling factorials in (5.5) gives

$$\mathcal{A} - 2F = \operatorname{diag}\left(\frac{1}{(2n_i+1)(2n_i+2)}\right) L(V_{5-i+\ell})_{0\le i,\ell\le 4} \operatorname{diag}\left(\frac{R_\ell!}{F_0(2\ell+2)!}\right),$$

where $L_{i,i-k} = d_{i,2k+1}$ is lower triangular. Therefore

$$\mathscr{P}_{[5]} = -\Gamma(b)\det(V_{i+j+1})_{0\le i,j\le 4},$$

with the explicit positive factor

$$\Gamma(b) = \kappa J(b)^4 E(b)^5 \prod_{i=0}^{4}(4n_i+2)^{\underline{2i}} \prod_{\ell=0}^{4}\prod_{r=1}^{10+2\ell}(2b-18+r),$$
$$\kappa = \frac{5!(10!)^5}{2!4!6!8!10!} = 14932726824960000000 > 10^{19}.$$

Every linear factor is at least b for $b \ge 20$, so $\Gamma(b) \ge \kappa b^{230}$.

At the actual doubled Bernoulli values, Euler's even-zeta formula and row/column reversal yield

$$|\mathscr{P}_{[5]}| = \frac{\Gamma(b)(4F_0)^5}{(2\pi)^{10(b-5)}}\det\big(\zeta(2b-18+2i+2j)\big)_{0\le i,j\le 4}.$$

The determinant on the right is a positive moment determinant on the points m^{-2}. Keeping the support points $1, 1/4, 1/9, 1/16, 1/25$ in Cauchy–Binet proves

$$\det\big(\zeta(2b-18+2i+2j)\big) \ge V\,120^{-2b+18},$$
$$V = \prod_{1\le r<s\le 5}(r^{-2}-s^{-2})^2 = \frac{2401}{36864000000000000} > 10^{-14}.$$

For $b \ge 30$, all nine actual variables have absolute value at most

$$T_b = \frac{8(2b)^{16}F_0}{(2\pi)^{2b-18}} \ge 1.$$

Thus $|\mathscr{P}_{[\le 4]}| \le Kb^{230}T_b^4$. After cancellation of the common powers of 2π, using $1 < 2\pi < 8$ gives

$$\frac{|\mathscr{P}_{[\le 4]}|}{|\mathscr{P}_{[5]}|} \le C\frac{b^{64}960^{2b}}{(2b-18)!}, \qquad C = \frac{K2^{142}}{\kappa 4^5 V 120^{18}} < 10^{251} < 10^{256}. \tag{5.12}$$

The powers here can be checked directly: the uncancelled power of 2π is $2b+22$, and $(2b)^{64}$ comes from T_b^4. The bounds $K < 10^{213}$, $2^{142} < 10^{43}$, $\kappa > 10^{19}$, and $V > 10^{-14}$ already give the stated coarse bound on C.

The remaining comparison is an exact integer inequality:

$$2^{384}\,10^{256}\,1920^{64}\,960^{3840} < 3822!. \tag{5.13}$$

It is verified by integer multiplication, with both sides explicitly specified. Moreover, for $b \ge 1920$,

$$\frac{(b+1)^{64}960^{2b+2}/(2b-16)!}{b^{64}960^{2b}/(2b-18)!} = \left(1+\frac{1}{b}\right)^{64}\frac{960^2}{(2b-16)(2b-17)} < \frac{3}{4}.$$

Indeed $(1+1/b)^{64} < 3$ and both denominator factors are at least $b \ge 1920 = 2\cdot 960$. It follows from (5.12)–(5.13) that the leading homogeneous term cannot be cancelled when $b \ge 1920$. Since $D(b) > 0$ there, (5.6) proves nonsingularity throughout that range.

5.5 The single missing matrix has a small exact certificate

The congruence covers every $6 \le b < 1920$ except $b = 962$. At this value define

$$S = \operatorname{diag}(2, 1, 64, 1/8, 2)\,\mathcal{A}\,\operatorname{diag}(1, 1, 1/32, 1/16, 1/512).$$

Thus $\det S = 2^{-13}\det\mathcal{A}$. Every coefficient of this affine matrix in $U_1, \dots, U_9$ has 2-adic valuation at least -3. Therefore its reduction modulo 2^{20} is determined by the inputs modulo 2^{23}.

The nine input residues, in increasing order of the index j, are

$$(660565, 5568257, 602509, 3180025, 6422085, 3501169, 2681469, 1596009, 6743605) \pmod{2^{23}}.$$

They are obtained by the elementary Bernoulli recurrence

$$(2m+1)(2B_{2m}) = 2m - 1 - \sum_{j=1}^{m-1}\binom{2m+1}{2j}(2B_{2j}), \qquad m \ge 1. \tag{5.14}$$

Every divisor $2m+1$ is odd. The recurrence is valid modulo every power of two and produces the residues through $m = 961$ without floating arithmetic. A separate exact rational Bernoulli evaluation agrees with all nine inputs.

Substitution in (5.5) gives the following matrix modulo $2^{20} = 1048576$:

$$S \equiv \begin{pmatrix} 917951 & 442211 & 476112 & 551248 & 968980 \\ 679904 & 45903 & 922072 & 654544 & 644616 \\ 797935 & 540216 & 1028224 & 572168 & 196164 \\ 491016 & 385277 & 695495 & 798931 & 823817 \\ 917289 & 206367 & 412992 & 727792 & 439172 \end{pmatrix} \pmod{1048576}.$$

The determinant of this displayed integer matrix is $524288 = 2^{19}$ modulo 2^{20}. Hence

$$v_2(\det S) = 19, \qquad v_2(\det\mathcal{A}) = 32, \qquad v_2(\det F) = 26, \qquad v_2(\rho_{5,962}) = 6.$$

The value $v_2(\det F) = 26$ follows immediately from (5.7): its only even factors beyond 2^{14} are $(b-2)^2 = 960^2$, of valuation 12. An independent check evaluates $\det\mathcal{A}/\det F$ directly as

a rational number; it gives the same valuation. The original integer numerator also satisfies $P_{10}(962; 2B_{1922}, \dots, 2B_{1906}) \equiv 65536 \pmod{131072}$.

This is a finite nonvanishing certificate, not a numerical approximation. It closes the only missing index below the sufficient threshold. Together with the preceding congruence and dominance arguments, it proves Theorem 5.1.

5.6 Ordinary convergent identities supplied by the completed family

For $b \geq 6$ and $0 \leq k < b$, let $\lambda_t^{(b,k)} = (M_{5,b}^{-1})_{kt}$, $0 \leq t < b$, and define the finite odd polynomial and its centered difference

$$K_{b,k}(u) = \sum_{t=0}^{b-1} \lambda_t^{(b,k)} u^{2b+2t+1}, \qquad Q_{b,k}(x) = K_{b,k}(x + \tfrac{1}{2}) - K_{b,k}(x - \tfrac{1}{2}).$$

The centered difference is precisely the coefficient transform (5.1), so its continued moment sum equals Z_k. For the ordinary formulation put

$$\eta_{s,0} = \frac{1}{s-1}, \qquad \eta_{s,j} = \frac{(s)_{2j-1} B_{2j}(1/2)}{(2j)!} \quad (j \geq 1), \qquad c_{b,\ell} = \sum_{j=0}^{\ell} \eta_{10,j} \eta_{2b,\ell-j}.$$

Let $T_{b,k}(x)$ be the polynomial consisting of the nonnegative powers in

$$Q_{b,k}(x) x^{-2b-8} \sum_{\ell \geq 0} c_{b,\ell} x^{-2\ell}.$$

Only finitely many coefficients enter it, and its degree is at most $2b - 10$. Then the completed family gives the exact identities

$$\boxed{\sum_{n=0}^{\infty} \left\{ Q_{b,k}(x_n) \frac{\psi^{(9)}(n+1)\psi^{(2b-1)}(n+1)}{9!\,(2b-1)!} - T_{b,k}(x_n) \right\} = \zeta(2k).} \tag{5.15}$$

The summands are $O(x_n^{-2})$. Every removed nonnegative even power has centered continued sum $\zeta(-2j, 1/2) = 0$, so the ordinary absolutely convergent sum has exactly the same value as the moment combination. Formula (5.15) is a finite rational algorithm at each b, k, with nonsingularity now proved for the entire parameter range. At $k = 0$ its value is the rational number $-1/2$.

Status and further research. The new logical step is the exact $b = 962$ certificate and the combination of the two independently audited ranges. The source congruence, the finite contraction, and the eventual dominance mechanism are retained with attribution. The same strategy is available for smaller order twelve, but its finite cases and modular determinant have not been proved here. A uniform argument for all unequal even orders, or even the full gap-six family, is still a substantive open problem. Higher-minor positivity cannot simply be inferred from the known pairwise ratio inequalities; the determinant sign already changes in some families.

Reproducible artifacts. `completion_certificate.py` proves the missing case from the displayed five-square matrix using no external project files. `verify_eventual_ten.py` rederives all 25 entry norms and the exact comparison (5.13). `verify_p10_dependencies.py`, its attributed algebra helper, and `p10_dependency_certificate.json` replay the full inherited polynomial identity, modular substitution, and all small base matrices. The three output records distinguish full replay from the deliberately optional quick mode. None uses floating-point evidence for a theorem.

6 Several transverse harmonic marks and Gamma–Stieltjes gaps

The harmonic-gap continuation [9] leaves a precise question after its single-marker theorem: can two independent transverse derivatives be taken at adjacent deleted indices, while retaining the prescribed relative continuation at nonintegral endpoints? We answer its first target, $\partial_u\partial_v\mathcal{H}_{a,b}(s,u,v,t)|_{u=v=0}$, and every analogous cluster of identical marks. A finite set-partition formula also determines the fully symmetric part of arbitrarily different marks. The partition coefficients are the classical symmetric-sum coefficients of Hoffman [7, 8]. The point requiring proof here is their local realization by decorated relative quotients, including products of Hurwitz primitives and their simultaneous resonant derivatives. An individual ordered word with unequal marks retains information that the symmetric formula does not determine.

6.1 Normalization and closure under finite decorations

Throughout this section, $\Re a, \Re b > 0$, and all powers use the analytic principal logarithm on the right half-plane. For sufficiently decaying lattice functions, define

$$Z_c[f_1,\dots,f_d] = \sum_{n_1>\cdots>n_d\ge 0}\prod_{j=1}^{d} f_j(n_j+c), \qquad Z_c[\varnothing]=1.$$

The relative quotient is fixed by

$$\mathcal{H}_{a,b}[w] = Z_b[w] - \sum_{j=1}^{|w|} Z_a[w_{\le j}]\mathcal{H}_{a,b}[w_{>j}], \qquad \mathcal{H}_{a,b}[\varnothing]=1. \tag{6.1}$$

For power letters $f_s(x)=x^{-s}$, write $\mathcal{H}_{a,b}(s_1,\dots,s_d)$. This is the relative interpolant already used in the source; no new endpoint interpolation is being chosen.

For clarity, its joint spectral entireness follows from the published Ihara–Nakamura–Yamamoto theory [6, Corollary 4.2 and Proposition 5.3]. With deconcatenation convolution denoted by $*$, put $\Phi_c(w)=\Psi(w;c-1)$ in their notation. Their formula is $\Phi_c = Z_c^{-1} * Z_1$; consequently

$$\mathcal{H}_{a,b} = Z_a^{-1} * Z_b = \Phi_a * (\Phi_b)^{-1}.$$

On each fixed word, convolution and inversion are finite polynomial operations in the entire Φ_c-values. The harmonic product law also follows from the character property [6, Theorem 5.2]. These identities initially hold on an open absolute-convergence domain and then continue analytically.

Use the completed Hurwitz function

$$Q(u,x) = \zeta(u,x) - \frac{1}{u-1}, \qquad Q(u,x+1) = Q(u,x) - x^{-u}. \tag{6.2}$$

It is entire in u and holomorphic for $\Re x>0$. Write $Q^{[r]}$ and $\zeta^{[r]}$ for derivatives in the first argument. The Stieltjes convention is

$$Q(1+\varepsilon,x) = \sum_{r\ge 0}\frac{(-1)^r\gamma_r(x)}{r!}\varepsilon^r, \qquad Q^{[r]}(1,x) = (-1)^r\gamma_r(x). \tag{6.3}$$

A decorated letter is x^{-s} multiplied by a polynomial in finitely many $Q(u,x)$ and $Q(u,x+1)$. Its Z_c and $\mathcal{H}_{a,b}$ are first defined by their actual convergent series and (6.1). A common right half-plane in all spectral variables exists.

Lemma 6.1 (One tail with arbitrary other decorations)**.** *Let $w = (f_1, \dots, f_d)$. Denote by $M_i(u)w$ the word with $f_i(x)$ replaced by $f_i(x)Q(u,x)$, by $I_j(u)w$ the insertion of f_u after position j, and by $C_j(u)w$ the replacement of f_j by $f_j f_u$. Then*

$$\mathcal{H}_{a,b}[M_i(u)w] = Q(u,a)\mathcal{H}_{a,b}[w] + \sum_{j=0}^{i-1} \mathcal{H}_{a,b}[I_j(u)w] + \sum_{j=1}^{i} \mathcal{H}_{a,b}[C_j(u)w]. \tag{6.4}$$

The other letters may already carry arbitrary finite decorations.

Proof. First replace Q by ζ and work where all relevant sums converge absolutely. In the infinite decorated sum, the additional Hurwitz-tail index is at least the index in position i. Its strict positions are the i slots preceding that position; its equality cases are the first i positions. This gives the insertion and merger terms with Z_c in place of $\mathcal{H}_{a,b}$, without an endpoint term.

Expand the relative recursion and group terms by their cut in the original word. At a cut before position i, the insertions and mergers contained in the prefix sum to $\zeta(u,a)Z_a[\text{prefix}]$, by the ordinary singleton stuffle identity. At a cut at or after position i, the same counting identity gives the decorated prefix. The remaining suffix terms are exactly the recursion for the right side of (6.4). Triangular uniqueness in the word length proves that formula with ζ. Subtracting $\mathcal{H}_{a,b}[w]/(u-1)$ gives the stated completed formula. This argument only uses pointwise products of the other letters, so it applies unchanged when they are decorated. □

Theorem 6.2 (Entire continuation of every finite decoration)**.** *Every decorated relative word above continues jointly entirely in all its spectral variables. It has a terminating finite expansion into bare relative words at affine sums of spectral arguments, with coefficients polynomial in entire endpoint functions $Q(U,a)$. All fixed independent spectral derivatives are legitimate.*

Proof. Use (6.2) to remove shifted factors. Apply Lemma 6.1 to one unshifted factor. Every resulting relative word contains one fewer Q factor: an insertion or merger adds a bare power, while the remaining decorations are unchanged. Induction ends with the entire bare relative words. Each step agrees with the initial convergent quotient, so the resulting entire expression is its analytic continuation. This proves both existence and independence of the elimination order. In the initial domain, Hurwitz asymptotics uniformly on a slightly larger spectral compact set justify termwise differentiation; elsewhere the derivatives follow from the proved entire identities. In particular the argument does not infer an analytic identity from equality at integer endpoint differences. □

6.2 The local symmetric-sum theorem

For a finite separable polynomial $P(y,z)$, define linearly

$$\mathcal{E}^{s,t}_{a,b}[A(y)B(z)] = \mathcal{H}_{a,b}[f_s A, f_t B]. \tag{6.5}$$

We use this map for polynomials in the completed gap primitives and their derivatives. Theorem 6.2 specifies its continuation. It also has an explicit ordinary-series meaning. With $F(y,z) = y^{-s}z^{-t}P(y,z)$,

$$\begin{aligned}\mathcal{E}^{s,t}_{a,b}[P] = &\sum_{n>j\ge 0} F(n+b, j+b) + \sum_{0\le n\le j} F(n+a, j+a) \\ &- \sum_{n,j\ge 0} F(n+a, j+b),\end{aligned} \tag{6.6}$$

whenever these sums converge absolutely. This follows by expanding the depth-two case of (6.1); the two a, a terms combine to the weak opposite ordering.

Put

$$g_u(y,z) = Q(u,z+1) - Q(u,y). \tag{6.7}$$

Taking the difference between the insertions at the two boundary letters in Lemma 6.1, and using (6.2), gives

$$\mathcal{E}^{s,t}_{a,b}[g_u] = \mathcal{H}_{a,b}(s,u,t). \tag{6.8}$$

The same assertion holds with arbitrary additional boundary decorations.

Lemma 6.3 (Products of gap primitives)**.** *For independent complex $u_1,\dots,u_k$, let $U_B = \sum_{i\in B} u_i$. Then*

$$\mathcal{E}^{s,t}_{a,b}\left[\prod_{i=1}^{k} g_{u_i}\right] = \sum_{(B_1,\dots,B_\ell)} \mathcal{H}_{a,b}(s,U_{B_1},\dots,U_{B_\ell},t), \tag{6.9}$$

where the sum ranges over all ordered set partitions of $\{1,\dots,k\}$. The identity is jointly entire.

Proof. After some marked positions have been inserted, write their variables as $x_1,\dots,x_\ell$, with $x_0 = y$ and $x_{\ell+1} = z$. The exact telescoping identity

$$g_v(y,z) = \sum_{h=0}^{\ell} g_v(x_h,x_{h+1}) + \sum_{h=1}^{\ell} x_h^{-v} \tag{6.10}$$

follows from (6.2). It is an identity of primitive functions, without an integer-lattice assumption on the external endpoints. By (6.8), the first sum inserts the next label in each possible strict slot, while the second merges it with each existing block. Starting with one marker, this constructs every ordered set partition once. All operations are identities between the initial convergent quotients; Theorem 6.2 supplies their joint continuation. □

For a set partition π of $\{1,\dots,k\}$, set

$$\mu(\pi) = \prod_{B\in\pi} (-1)^{|B|-1}(|B|-1)!.$$

Theorem 6.4 (Local symmetric-sum identity)**.** *For arbitrary complex spectral arguments,*

$$\sum_{\sigma\in S_k} \mathcal{H}_{a,b}(s,u_{\sigma(1)},\dots,u_{\sigma(k)},t) = \mathcal{E}^{s,t}_{a,b}\left[\sum_{\pi} \mu(\pi) \prod_{B\in\pi} g_{U_B}(y,z)\right]. \tag{6.11}$$

Proof. Apply Möbius inversion on the lattice of equality partitions to (6.9). A block B has Möbius coefficient $(-1)^{|B|-1}(|B|-1)!$. This is the classical symmetric-sum argument [7, 8]; Lemma 6.3 proves that it acts on the present local analytic expressions. For example, at $k=2$,

$$\mathcal{H}(s,u,v,t) + \mathcal{H}(s,v,u,t) = \mathcal{E}[g_u g_v - g_{u+v}].$$

Every function used in the inversion has already been continued jointly entirely. □

6.3 Independent derivatives and identical marked clusters

For integers $r_i \geq 0$ and centers $\lambda_i \in \mathbb{C}$, write

$$r_B = \sum_{i\in B} r_i, \qquad \lambda_B = \sum_{i\in B} \lambda_i, \qquad p_B(y,z) = Q^{[r_B]}(\lambda_B, z+1) - Q^{[r_B]}(\lambda_B, y).$$

Corollary 6.5 (All independently marked symmetric sums)**.** *The exact identity*

$$\prod_{i=1}^{k} \partial_{u_i}^{r_i} \sum_{\sigma\in S_k} \mathcal{H}_{a,b}(s,u_{\sigma(1)},\dots,u_{\sigma(k)},t)\Big|_{u_i=\lambda_i} = \mathcal{E}_{a,b}^{s,t}\Bigg[\sum_{\pi}\mu(\pi)\prod_{B\in\pi} p_B(y,z)\Bigg] \tag{6.12}$$

uses at most 2^k-1 *different gap primitives. At every merged resonance* $\lambda_B=1$*, the corresponding primitive is*

$$p_B(y,z) = (-1)^{r_B}\big(\gamma_{r_B}(z+1)-\gamma_{r_B}(y)\big). \tag{6.13}$$

Proof. Differentiate the entire identity (6.11). Within a block, every variable enters U_B with coefficient one, so its derivative order is exactly r_B, without an extra multinomial coefficient. Equation (6.3) proves (6.13). Simultaneous resonances therefore require no directional limit or separately chosen finite part. □

When all pairs (λ_i,r_i) equal (λ,r), each permutation term is identical. Put

$$\begin{aligned} p_j &= Q^{[jr]}(j\lambda,z+1)-Q^{[jr]}(j\lambda,y),\\ e_k(p_1,\dots,p_k) &= [v^k]\exp\left(\sum_{j=1}^{k}\frac{(-1)^{j-1}}{j}p_jv^j\right). \end{aligned} \tag{6.14}$$

The set-partition formula becomes the individual ordered identity

$$\partial_{u_1}^{r}\cdots\partial_{u_k}^{r}\mathcal{H}_{a,b}(s,u_1,\dots,u_k,t)\Big|_{u_1=\cdots=u_k=\lambda} = \mathcal{E}_{a,b}^{s,t}[e_k(p_1,\dots,p_k)]. \tag{6.15}$$

These are independent derivatives taken before specialization. A total derivative in one common variable would give a different sum. The coefficient in (6.14) is a finite polynomial operation; no convergence of an all-k marked generating series is asserted. The first useful polynomials are

$$\begin{aligned} e_2 &= \tfrac{1}{2}(p_1^2-p_2),\\ e_3 &= \tfrac{1}{6}(p_1^3-3p_1p_2+2p_3),\\ e_4 &= \tfrac{1}{24}(p_1^4-6p_1^2p_2+3p_2^2+8p_1p_3-6p_4). \end{aligned} \tag{6.16}$$

6.4 Explicit Gamma and Stieltjes identities

For zero-centered first derivatives define

$$L(y,z)=\log\Gamma(z+1)-\log\Gamma(y),\qquad R_j(y,z)=\zeta^{[j]}(0,z+1)-\zeta^{[j]}(0,y).$$

The classical identity $\zeta^{[1]}(0,x)=\log\Gamma(x)-\frac{1}{2}\log(2\pi)$ [4] makes $p_1=L$, while $p_j=R_j$ for $j\ge 2$. The source's first two-marker target is therefore

$$\boxed{\partial_u\partial_v\mathcal{H}_{a,b}(s,u,v,t)\big|_{u=v=0}=\frac{1}{2}\mathcal{E}_{a,b}^{s,t}[L^2-R_2].} \tag{6.17}$$

This is an identity at arbitrary independent s,t and arbitrary right-half-plane endpoints. It supplies ordinary absolutely convergent sums through (6.6) whenever $\Re s,\Re t>3$: $\log\Gamma(x)=O(x\log x)$ and $\zeta^{[2]}(0,x)=O(x\log^2 x)$ on each fixed lattice give summability of every separated letter. Elsewhere it is the specified entire continuation. The third and fourth identical marked

clusters are obtained from (6.16); no new ordered coordinate is required for these symmetric identical letters.

To connect this directly with normalized antiderivatives, write the Hurwitz parameter-derivative law in its entire form

$$\partial_x \zeta(u,x) = -1 - uQ(u+1,x).$$

Differentiating at $u=0$, for every $j \geq 1$, gives

$$\partial_x \zeta^{[j]}(0,x) = (-1)^j j \gamma_{j-1}(x), \qquad R_j(y,z) = (-1)^j j \int_y^{z+1} \gamma_{j-1}(x)\,dx. \tag{6.18}$$

The paths lie in the right half-plane; the constants are fixed by vanishing when $z+1=y$. Thus (6.17) is also

$$\partial_u \partial_v \mathcal{H}_{a,b}(s,u,v,t)|_{0,0} = \frac{1}{2}\mathcal{E}^{s,t}_{a,b}\Big[L(y,z)^2 - 2\int_y^{z+1} \gamma_1(x)\,dx\Big]. \tag{6.19}$$

The logarithm in L remains the difference of analytic log-Gamma branches; it is not replaced by an independently chosen logarithm of a ratio.

An example in which the merging itself creates a Stieltjes coefficient uses two half-order marks. Set $A(y,z) = \zeta^{[1]}(1/2, z+1) - \zeta^{[1]}(1/2, y)$. Then

$$\partial_u \partial_v \mathcal{H}_{a,b}(s,u,v,t)|_{u=v=1/2} = \frac{1}{2}\mathcal{E}^{s,t}_{a,b}[A(y,z)^2 - \gamma_2(z+1) + \gamma_2(y)]. \tag{6.20}$$

Indeed the merged index is one and its total derivative order is two.

For an unequal pair, the symmetric distinction is essential. Define

$$J_{0,1}(s,t) = \partial_u \partial_v \{\mathcal{H}_{a,b}(s,u,v,t) + \mathcal{H}_{a,b}(s,v,u,t)\}\big|_{u=0,v=1}.$$

The exact mixed Gamma–Stieltjes formula is

$$J_{0,1}(s,t) = \mathcal{E}^{s,t}_{a,b}[-L(y,z)\{\gamma_1(z+1) - \gamma_1(y)\} - \gamma_2(z+1) + \gamma_2(y)]. \tag{6.21}$$

This determines the sum of the two ordered derivatives, not each summand separately.

6.5 Several gaps and the exact scope of the reduction

For surviving variables $x_1, \ldots, x_d$, the three types of completed gaps are

$$\begin{cases} Q(u, x_1+1) - Q(u,a), & \text{above the first variable,} \\ Q(u, x_{j+1}+1) - Q(u, x_j), & \text{between successive variables,} \\ Q(u,b) - Q(u, x_d), & \text{below the last variable.} \end{cases} \tag{6.22}$$

On an absolute-convergence domain, Z_c is a quasi-shuffle character on lattice functions with pointwise multiplication. Characters with values in a commutative algebra form a group under deconcatenation convolution; therefore $Z_a^{-1} * Z_b$ has this property for decorated letters as well. The continuation proved above preserves these identities. The upper boundary in (6.22) follows from the one-tail lemma at the first position and the shift law. The lower boundary follows by subtracting that lemma at the last position from the singleton stuffle identity, using $\mathcal{H}_{a,b}(u) = Q(u,b) - Q(u,a)$.

Apply the partition polynomial independently in each gap and multiply the resulting polynomials before expanding into decorated letters. The proof by insertion applies gap by gap, and Theorem 6.2 gives joint continuation, including all shared spectral resonances. It

determines the ordered word whenever each gap contains one mark or an identical marked cluster. With no surviving variables, the sole gap is $Q(u,b)-Q(u,a)$, and the construction reduces to the classical symmetric-sum formula for the relative interpolant.

The accompanying program verifies the local symmetric identity through five independent labels by exact integer arithmetic on formal decorated words. All endpoint factors and shorter words cancel, leaving precisely the $k!$ permutations. A separate numerical check uses the original nested Z_c-series and relative quotient at nonintegral complex endpoints: increasing the cutoff from 80 to 160 terms reduces its absolute residual from 1.66×10^{-36} to 2.69×10^{-40}. An independent reconstruction of the actual zero-marker derivative at those noninteger endpoints, using convergent nested sums and a symmetrized top-tail continuation, gives residuals 1.40×10^{-27} and 3.73×10^{-31} at the same cutoffs. Individual singular double-zeta terms are never differentiated at $(0,0)$. Five direct finite-interval logarithmic comparisons test negative centers, the Gamma square, three marks, and merged Stieltjes resonances at 60 decimal digits; each residual is below 8×10^{-62}. These are diagnostics of the algebra and normalization, not interval certificates or substitutes for the analytic proof.

The audited source correctly withheld the several-marker claim; no correction to its single-marker theorem is needed. The present theorem answers its first Q4 target and the fully symmetric part of Q5. A remaining precise question is to represent the antisymmetric ordered two-marker component with unequal marks. Another is to classify which unmarked nonpositive deletions preserve this symmetric span: such deletions can shift different marked centers unequally, so their full ordered extension is not implied by the Bell formula.

7 Higher spectral derivatives through the faces of a simplex

The incoming beta-polarization report asks for explicit higher-order subtractions on a Dirichlet simplex and for their extension to odd polynomial weights [10, Further research, items 3–4]. The incoming quartic report already gives a finite-subtraction compiler at every spectral order [2, Section 5]. We give a different answer: a finite representation by ordinary simplex integrals of *lower* one-mass derivatives. Its useful structural feature is that an interaction involving a face with k vertices cannot appear before the kth spectral derivative, after an explicitly computable zero-jet polynomial has been separated.

The scalar Dirichlet parameter identity is classical. It is already used for mixed twists in the canonical manuscript [1, `07-mixed-twist-identities.tex`]; logarithmic Dirichlet averages have a substantial classical theory [11, 12]. The contribution here is the explicit face subtraction, its absolute-convergence proof, the resulting all-order formula, and the rational zero-jet term. This is a claim of new content relative to the inspected repository, not an assertion of exhaustive literature priority.

7.1 Spectral normalization and the zero-jet polynomial

Fix $a>0$, $r\ge2$, real masses $B_1,\dots,B_r\ge0$, and a polynomial $W(x)=\sum_{d=0}^{D}w_dx^d$. Write $x_n=n+a$ and initially define

$$E_W(s;B)=\sum_{n\ge0}W(x_n)(x_n^2+B)^{-s}, \tag{7.1}$$

$$Z_{W,\boldsymbol{B}}(s)=\sum_{n\ge0}W(x_n)\prod_{i=1}^{r}(x_n^2+B_i)^{-s}, \tag{7.2}$$

$$\mathcal{D}_W(s)=\frac{1}{r}\sum_{i=1}^{r}E_W(rs;B_i)-Z_{W,\boldsymbol{B}}(s),\qquad \Delta_m(W)=\mathcal{D}_W^{(m)}(0). \tag{7.3}$$

The sums start in their half-planes of absolute convergence and are then continued meromorphically. All logarithms of the positive quadratic factors are real. Let

$$F_j(B) = \partial_s^j E_W(s;B)\Big|_{s=0}. \tag{7.4}$$

Lemma 7.1 (The polynomial at spectral order zero)**.** *The spectral functions in* (7.1)–(7.2) *are holomorphic at zero. The one-mass zero jet is*

$$F_0(B) = \sum_{d=0}^{D} w_d \zeta(-d,a) + \sum_{h\ge 1} p_h B^h, \qquad p_h = \frac{(-1)^h}{2h}\, w_{2h-1}, \tag{7.5}$$

where missing polynomial coefficients are zero. In particular F_0 is mass-independent for even W.

Proof. For sufficiently large x, expand the quadratic factor with a finite remainder:

$$(x^2+B)^{-s} = x^{-2s} \sum_{h=0}^{K} \frac{(-1)^h (s)_h}{h!} B^h x^{-2h} + \text{remainder}.$$

After multiplication by x^d, the removed terms sum to $(-1)^h (s)_h B^h \zeta(2s+2h-d,a)/h!$. Choose K large enough that the remainder and its required derivatives converge normally near zero. A finite head is retained exactly, so no restriction such as $B < a^2$ is needed. For $h \ge 1$, the coefficient vanishes at zero unless $2h-d=1$. In that case

$$\frac{(s)_h}{h!} = \frac{s}{h} + O(s^2), \qquad \zeta(1+2s,a) = \frac{1}{2s} + O(1),$$

which gives the term $(-1)^h B^h/(2h)$ in (7.5). The same subtraction for the product of quadratic factors has positive Taylor-degree coefficients vanishing at $s=0$; they cancel its possible simple Hurwitz poles. This also proves holomorphy of the product spectrum. The argument is locally uniform in the masses and in $a>0$. □

7.2 An explicit subtraction on every face

For a nonempty set $S \subseteq \{1,\dots,r\}$, let

$$\Sigma_S = \{(t_i)_{i\in S} : t_i \ge 0,\ \sum_{i\in S} t_i = 1\}.$$

The measure is ordinary $(|S|-1)$-dimensional coordinate measure: $dt_1 \cdots dt_{k-1}$ after the final coordinate has been eliminated. Thus the simplex has volume $1/(k-1)!$, where $k=|S|$. For a nonempty $T \subseteq S$ put

$$t_T = \sum_{i\in T} t_i, \qquad B_T = \frac{\sum_{i\in T} t_i B_i}{t_T}, \qquad G_T(F;t) = t_T F(B_T).$$

Set $G_T = 0$ when $t_T = 0$. Define the face subtraction

$$\boxed{\mathcal{H}_S(F;t) = \sum_{\varnothing \ne T \subseteq S} (-1)^{|S|-|T|}\, t_T F(B_T).} \tag{7.6}$$

For a pair this is the difference from the affine chord:

$$\mathcal{H}_{ij}(F;u) = F(uB_i + (1-u)B_j) - uF(B_i) - (1-u)F(B_j). \tag{7.7}$$

For three vertices it is

$$\mathcal{H}_{ijk}(F;t) = F(t_iB_i + t_jB_j + t_kB_k) + \sum_{\ell\in\{i,j,k\}} t_\ell F(B_\ell) - \sum_{\{\ell,m\}\subseteq\{i,j,k\}} (t_\ell + t_m)F\Big(\frac{t_\ell B_\ell + t_m B_m}{t_\ell + t_m}\Big). \tag{7.8}$$

These formulas remain regular when masses coincide. They use neither division by a mass difference nor an unspecified finite part.

Lemma 7.2 (Boundary cancellation and ordinary integrability)**.** *If $F \in C^1([\min B_i, \max B_i])$, then $\mathcal{H}_S$ vanishes on every proper coordinate face and*

$$|\mathcal{H}_S(F;t)| \le C \min_{i\in S} t_i. \tag{7.9}$$

For $k = |S| \ge 2$, the integral

$$\int_{\Sigma_S} \mathcal{H}_S(F;t) \prod_{i\in S} t_i^{s-1}\, dt \tag{7.10}$$

is absolutely convergent for $\operatorname{Re} s > -1/(k-1)$. It is holomorphic there, with locally normally convergent fixed logarithmic derivatives. The same conclusions hold for a holomorphic family of F whose C^1 norms are locally bounded.

Proof. The perspective $G_T = t_T F(B_T)$ is Lipschitz on the nonnegative coordinate cone. Away from $t_T = 0$,

$$\partial_{t_i} G_T = F(B_T) + F'(B_T)(B_i - B_T) \quad (i \in T),$$

so its coordinate derivatives have a uniform bound. Continuity at the origin extends the Lipschitz bound there. If $t_i = 0$, the terms indexed by T and $T \cup \{i\}$ cancel in (7.6). Changing that coordinate from zero to t_i therefore gives a bound Ct_i for every i, proving (7.9).

Cover the simplex by the k regions where a specified coordinate is at least $1/k$. On such a region,

$$\min_i t_i \le \Bigg(\prod_{i\ne i_{\max}} t_i\Bigg)^{1/(k-1)}.$$

The remaining coordinate factors are dominated by a product of integrable powers whenever $\operatorname{Re} s + 1/(k-1) > 0$. Any fixed power of $\sum_i |\log t_i|$ remains integrable on smaller closed half-planes. Dominated holomorphic parameter integration proves all assertions. In particular, every integral below with the weight $\prod_i t_i^{-1}$ is an ordinary absolutely convergent integral. □

Theorem 7.3 (Exact decomposition by simplex faces)**.** *Put*

$$Q_k(s) = \frac{\Gamma(1+ks)}{\Gamma(1+s)^k}. \tag{7.11}$$

The following identity holds meromorphically for $\operatorname{Re} s > -1/(r-1)$, with ordinary face integrals away from spectral poles, and in particular as a holomorphic germ at zero:

$$\boxed{\mathcal{D}_W(s) = -\frac{1}{r}\sum_{k=2}^{r} s^{k-1} Q_k(s) \sum_{|S|=k} \int_{\Sigma_S} \mathcal{H}_S(E_W(rs;\cdot);t) \prod_{i\in S} t_i^{s-1}\, dt.} \tag{7.12}$$

Proof. The product of Gamma integrals, with variables changed to a radial coordinate and simplex proportions, gives

$$\prod_{i=1}^{r} y_i^{-s} = \frac{\Gamma(rs)}{\Gamma(s)^r} \int_{\Sigma_{\{1,\dots,r\}}} \left(\sum_i t_i y_i \right)^{-rs} \prod_i t_i^{s-1}\, dt$$

for $y_i > 0$ and $\operatorname{Re} s > 0$. Apply this to $y_i = x_n^2 + B_i$ and sum where the spectra converge absolutely. Finite subset inversion gives

$$F\left(\sum_i t_i B_i \right) = \sum_{\varnothing \neq S \subseteq \{1,\dots,r\}} \mathcal{H}_S(F;t).$$

On the full simplex, each $\mathcal{H}_S$ is homogeneous of degree one in the coordinates belonging to S. Under the normalized symmetric Dirichlet weight, $q = t_S$ has mean $|S|/r$ and is independent of the normalized proportions t_i/q, $i \in S$. Equivalently, this follows by integrating first in q and the complementary proportions using the ordinary beta integral. Therefore, for $k = |S|$,

$$\mathbb{E}_{r,s} \mathcal{H}_S(F;t) = \frac{k}{r} \frac{\Gamma(ks)}{\Gamma(s)^k} \int_{\Sigma_S} \mathcal{H}_S(F;t) \prod_{i \in S} t_i^{s-1}\, dt.$$

The singleton terms are $r^{-1} \sum_i F(B_i)$. Subtracting them and using $\frac{k}{r}\Gamma(ks)/\Gamma(s)^k = s^{k-1} Q_k(s)/r$ proves (7.12) in the initial half-plane. The spectral finite subtractions in Lemma 7.1, together with Lemma 7.2, give meromorphic continuation of the integrals and the asserted holomorphic germs. The equality follows there by continuation from the common convergence domain. □

7.3 The entire zero-jet contribution is a finite rational function

Separate $E_W(rs;B) = F_0(B) + (E_W(rs;B) - F_0(B))$ in (7.12). Only the mass polynomial in (7.5) contributes to the first part.

Theorem 7.4 (Rational zero-jet contribution)**.** *The part of $\mathcal{D}_W(s)$ arising from F_0 is*

$$\boxed{\mathcal{L}_W(s) = \sum_{h \geq 1} p_h \left\{ \frac{1}{r} \sum_{i=1}^{r} B_i^h - \frac{h!}{(rs)_h} \sum_{h_1 + \cdots + h_r = h} \prod_{i=1}^{r} \frac{(s)_{h_i} B_i^{h_i}}{h_i!} \right\}.} \tag{7.13}$$

All apparent singularities at $s = 0$ are removable, and $\mathcal{L}_W(0) = 0$. The term $h = 1$ vanishes identically. For $m \geq 1$, set

$$C_{h,j} = [z^h] \prod_{i=1}^{r} (1 - B_i z)^{j/r}, \qquad 1 \leq j \leq h-1. \tag{7.14}$$

Then

$$\boxed{\mathcal{L}_W^{(m)}(0) = \sum_{h \geq 2} p_h (-1)^{m+1} m! r^m h! \sum_{j=1}^{h-1} \frac{(-1)^j C_{h,j}}{j!(h-1-j)!\, j^{m+1}}.} \tag{7.15}$$

The coefficient extraction in (7.14) is finite and algebraic; it makes no branch choice beyond the formal series with constant coefficient one.

Proof. Dirichlet moments give

$$\mathbb{E}_{r,s} \left(\sum_i t_i B_i \right)^h = \frac{h!}{(rs)_h} \sum_{h_1 + \cdots + h_r = h} \prod_i \frac{(s)_{h_i} B_i^{h_i}}{h_i!}.$$

This proves (7.13). At $s = 0$ its continuation is the average of the vertex values: only compositions supported on a single index survive after the common factor s has been cancelled.

For the derivative formula put $v = rs$. The hth moment becomes

$$M_h(v) = \frac{h!}{(v)_h}[z^h]\prod_i (1 - B_i z)^{-v/r}.$$

After cancellation at zero, it is a rational function of numerator and denominator degree at most $h - 1$. Its polynomial part is constant. Its possible poles are simple and lie at $v = -j$, $1 \le j < h$, with residues

$$\operatorname{Res}_{v=-j} M_h(v) = \frac{(-1)^j h! C_{h,j}}{j!(h-1-j)!}.$$

Differentiate the partial fractions m times at $v = rs = 0$ and subtract from the vertex average. This is exactly (7.15). □

For example, let $V(\boldsymbol{B}) = \sum_{i<j}(B_i - B_j)^2$. For the first nontrivial odd weight,

$$\mathcal{L}_{x^3}(s) = \frac{sV(\boldsymbol{B})}{4r(1+rs)}, \qquad \mathcal{L}^{(m)}_{x^3}(0) = \frac{m!(-r)^{m-1}}{4r}V(\boldsymbol{B}). \tag{7.16}$$

More generally, after division by $m!$, the sequence in (7.15) is a finite linear combination of powers $(-r/j)^m$. It therefore satisfies an explicit finite recurrence in m. This rational contribution is the part supplied by the zero one-mass jet. It should not be read as saying that Stieltjes or residue terms inside the lower jets $F_1, F_2, \ldots$ disappear.

7.4 Every derivative uses only finitely many lower jets

For $k \ge 2$ and $q \ge 0$ define

$$P_{k,q}(L) = [z^q]\exp\left(Lz + \sum_{n\ge 2}\frac{(-1)^n\zeta(n)}{n}(k^n - k)z^n\right). \tag{7.17}$$

Only finitely many terms of the exponential are required at any fixed q. In particular,

$$P_{k,0} = 1, \qquad P_{k,1} = L, \qquad P_{k,2} = \frac{L^2}{2} + \frac{k^2 - k}{2}\zeta(2).$$

Put $\ell_S(t) = \sum_{i\in S}\log t_i$.

Theorem 7.5 (All-order formula and face-size restriction)**.** *For every integer $m \ge 1$,*

$$\boxed{\begin{aligned}\Delta_m(W) &= \mathcal{L}^{(m)}_W(0)\\ &\quad - \frac{m!}{r}\sum_{k=2}^{\min(r,m)}\sum_{|S|=k}\sum_{j=1}^{m-k+1}\frac{r^j}{j!}\int_{\Sigma_S}\mathcal{H}_S(F_j;t)P_{k,m-k+1-j}(\ell_S(t))\frac{dt}{\prod_{i\in S}t_i}.\end{aligned}} \tag{7.18}$$

Every displayed simplex integral is absolutely convergent. After the explicit zero-jet term is removed, a k-vertex face starts at order s^k and uses one-mass derivatives of order at most $m - k + 1$ in the mth derivative.

Proof. The classical Taylor series of $\log\Gamma(1+z)$ gives

$$Q_k(s) = \exp\left(\sum_{n\ge 2}\frac{(-1)^n\zeta(n)}{n}(k^n - k)s^n\right);$$

the Euler-constant terms cancel. Once F_0 has been removed, the inner function in (7.12) is $\sum_{j\ge1} r^j s^j F_j/j!$. Multiplication by s^{k-1} proves that this face starts at degree k. Extract the coefficient of s^m from (7.12); the remaining exponential is precisely (7.17). Lemma 7.2 justifies extraction under each integral. Multiplication by $m!$ gives the theorem. □

For a compact statement of the first instances write

$$I_S(F)=\int_{\Sigma_S}\mathcal{H}_S(F;t)\frac{dt}{\prod_i t_i},\qquad J_S(F)=\int_{\Sigma_S}\mathcal{H}_S(F;t)\ell_S(t)\frac{dt}{\prod_i t_i},\qquad L_m=\mathcal{L}_W^{(m)}(0).$$

All subset sums below use distinct labels. The complete second and third derivatives are

$$\Delta_2(W)=L_2-2\sum_{|S|=2}I_S(F_1), \tag{7.19}$$

$$\Delta_3(W)=L_3-3r\sum_{|S|=2}I_S(F_2)-6\sum_{|S|=2}J_S(F_1)-6\sum_{|S|=3}I_S(F_1). \tag{7.20}$$

The fourth derivative is

$$\begin{aligned}\Delta_4(W)=L_4&-4r^2\sum_{|S|=2}I_S(F_3)-12r\sum_{|S|=2}J_S(F_2)\\&-24\sum_{|S|=2}\int_{\Sigma_S}\mathcal{H}_S(F_1;t)\left(\frac{\ell_S(t)^2}{2}+\zeta(2)\right)\frac{dt}{\prod_i t_i}\\&-12r\sum_{|S|=3}I_S(F_2)-24\sum_{|S|=3}J_S(F_1)-24\sum_{|S|=4}I_S(F_1).\end{aligned} \tag{7.21}$$

A subset sum is empty when its size exceeds r. The term $\zeta(2)$ in the pair integral of (7.21) comes from Q_2 and is part of the normalization. In particular the third derivative needs only pair integrals of the second one-mass jet and pair and triple integrals of the first jet.

7.5 Explicit Gamma integrands and a nonzero face family

For $W=1$, first-derivative factorization gives the classical expression

$$F_1(B)=\log\Gamma(a+i\sqrt{B})+\log\Gamma(a-i\sqrt{B})-\log(2\pi), \tag{7.22}$$

using the analytic logarithm of Gamma in the right half-plane. The corresponding even-weight expression, inherited from first-derivative factorization, is

$$F_1(B)=\sum_{\varepsilon=\pm1}\sum_{j=0}^{2d}\binom{2d}{j}(-\varepsilon i\sqrt{B})^{2d-j}\zeta'(-j,a+\varepsilon i\sqrt{B}),\qquad W=x^{2d}. \tag{7.23}$$

Thus the largest face size $k=m$ in (7.18) requires only ordinary Gamma at $W=1$, and a finite Hurwitz-derivative expression for every even polynomial weight. Its coefficients and integration kernels are fully explicit.

The following identity shows that the higher faces are substantive terms of this particular representation.

Proposition 7.6 (A factorial identity for every face size)**.** *For $k=|S|\ge2$,*

$$\boxed{I_S(B\mapsto B^2)=(-1)^{k-1}(k-2)!\sum_{\{i,j\}\subseteq S}(B_i-B_j)^2.} \tag{7.24}$$

Consequently, if $B_i=B+\epsilon\beta_i$ and F is C^3 near B,

$$I_S(F)=\frac{\epsilon^2F''(B)}{2}(-1)^{k-1}(k-2)!\sum_{\{i,j\}\subseteq S}(\beta_i-\beta_j)^2+O(\epsilon^3). \tag{7.25}$$

Proof. On the nonnegative coordinate cone the quadratic perspective has the variance decomposition

$$t_T B_T^2 = \sum_{i\in T} t_i B_i^2 - \sum_{\{i,j\}\subseteq T} \frac{t_i t_j}{t_T}(B_i - B_j)^2.$$

The first sum is annihilated by the face subtraction. Writing $t_T^{-1} = \int_0^\infty e^{-vt_T}\,dv$ in the second gives

$$\mathcal{H}_S(B^2;t) = (-1)^{k-1} \sum_{\{i,j\}\subseteq S} (B_i - B_j)^2 t_i t_j \times \int_0^\infty e^{-v(t_i+t_j)} \prod_{\ell\in S\setminus\{i,j\}} (1 - e^{-vt_\ell})\,dv.$$

Since $\mathcal{H}_S$ is homogeneous of degree one, radial integration shows that I_S equals the integral of $e^{-\sum_i t_i}\mathcal{H}_S(t)/\prod_i t_i$ on the positive orthant. The radial factor is $\int_0^\infty e^{-\rho}\,d\rho = 1$. After the displayed sign is removed, every term is nonnegative, so Tonelli's theorem applies. The two distinguished coordinates give $(1+v)^{-2}$, and each other coordinate gives

$$\int_0^\infty e^{-t}\frac{1-e^{-vt}}{t}\,dt = \log(1+v).$$

The remaining integral is

$$\int_0^\infty \frac{\log^{k-2}(1+v)}{(1+v)^2}\,dv = (k-2)!.$$

This proves (7.24). Taylor-expand $F(B+\epsilon\beta)$ through degree two. The constant and affine parts are annihilated; Lemma 7.2 controls the integrated remainder uniformly, proving (7.25). □

For (7.22),

$$F_1''(B) = \sum_{n\ge 0} \frac{1}{(x_n^2 + B)^2} > 0.$$

Hence every face size has a nonzero quadratic collision coefficient when the perturbations are not all equal. This does not prove irreducibility into any other chosen function class. In particular, $\mathcal{H}_S$ need not vanish merely because two of several masses coincide; "face contribution" is the precise description used here.

7.6 Periodic weights and cancellation of the mean

Corollary 7.7 (Mean-zero periodic weights remove the zero-jet term)**.** *Let c_n be q-periodic, and insert c_n in every spectral sum. Theorems 7.3 and 7.5 still hold. The rational zero-jet contribution is*

$$\mathcal{L}_{c,W}(s) = \overline{c}\,\mathcal{L}_W(s), \qquad \overline{c} = \frac{1}{q}\sum_{v=0}^{q-1} c_v. \tag{7.26}$$

Thus $\overline{c} = 0$ removes this term for every *polynomial weight, including odd weights.*

Proof. The auxiliary Dirichlet series is

$$\sum_{n\ge 0} c_n (n+a)^{-z} = q^{-z}\sum_{v=0}^{q-1} c_v \zeta\left(z, \frac{a+v}{q}\right).$$

Its residue at $z=1$ is $\overline{c}$. Repeat the proof of Lemma 7.1; the zero jet becomes

$$F_{c,0}(B) = \sum_d w_d q^d \sum_v c_v \zeta\left(-d, \frac{a+v}{q}\right) + \overline{c}\sum_{h\ge 1} p_h B^h.$$

The mass-independent part is annihilated by every face subtraction. All other steps use linearity and normal convergence; periodic weights are bounded and cause no additional convergence issue. □

For $W = 1$ the first jet has the explicit finite dictionary

$$F_{c,1}(B) = \sum_{v=0}^{q-1} c_v \Bigg\{ \log\Gamma\left(\frac{a+v+i\sqrt{B}}{q}\right) + \log\Gamma\left(\frac{a+v-i\sqrt{B}}{q}\right) - \log(2\pi) - 2\log q\left(\frac{1}{2} - \frac{a+v}{q}\right)\Bigg\}. \tag{7.27}$$

This follows by scaling each residue class to $q^{-2s}E_1(s; B/q^2)$ with shift $(a+v)/q$, where the subscript 1 here denotes the polynomial weight. Formula (7.27) therefore supplies ordinary Gamma integrands for alternating weights and character projections. The mass-independent constants can be omitted inside $\mathcal{H}_S$, but are retained in the full first-jet identity.

7.7 Independent evaluation and research consequences

For $a = 2$, $W = 1$, $r = 3$, and masses $(B_1, B_2, B_3) = (0.2, 0.7, 1.3)$, the directly evaluated spectral derivative is

$$\Delta_3 = -1.89996732348926709161224369129247092227472402 34\ldots. \tag{7.28}$$

One calculation used a finite ordinary spectral sum and an independently summed binomial–Hurwitz tail. Changing the head length and tail degree from $(55, 13)$ to $(85, 17)$ changed the result by 3.04×10^{-46}. A second calculation evaluated (7.20): the pair terms used convergent one-mass expansions, while the triple integral used Gamma values from (7.22). The errors in the full face formula at Gauss–Legendre orders 16, 24, and 32, with 45-digit arithmetic, were respectively 1.62×10^{-18}, 2.54×10^{-27}, and 8.20×10^{-32}. The last error is limited by the retained one-mass expansion. The triple integral was

$$I_{\{1,2,3\}}(F_1) = 0.05646009472353840494615376709290125 15\ldots.$$

For stable numerical integration the simplex was divided into its six coordinate orderings. On each, use $t = (1, u, uv)/(1 + u + uv)$; the weighted measure becomes $dt/(t_1t_2t_3) = du\,dv/(uv)$, and the face cancellation removes the corner singularity. The code and precision-dependent outputs are included with the article. These are numerical diagnostics, not interval enclosures or replacements for the proofs.

The results close the explicit face-subtraction problem and supply a complete odd-weight zero-jet correction in this lower-jet framework. Further concrete questions are to evaluate special Gamma face integrals in a fixed polylogarithmic function class; to obtain comparable multivariable formulas when the spectral orders of the factors vary independently; and to derive closed all-order collision coefficients for the individual face integrals beyond (7.25). The theorem itself does not require such reductions and makes no claim that the unresolved Gaussian period conjectures are settled.

8 Source audit, exact verification, and integration decisions

8.1 What changes in the source status

The article supplies advances at specific research interfaces. It does not replace the source's definitions or silently promote other open claims. The following status changes are proposed for integration.

Source target	Proposed status after review of this article
Quartic report, R4: fifth- and higher-power cancellation	The universal recurrence and anchored generator are proved in Theorems 2.3 and 2.4. Exact eight- and eleven-coordinate rows are supplied for powers five and six. Theorem 2.5 treats every repeated even order. The minimal general coordinate count remains a separate question.
Quartic report, R5: resonant parity	The complete Laurent normalization and all coefficients are proved in Theorem 4.1. The local spectral Taylor series has exact radius two. A uniform finite arithmetic reduction as the resonance index varies is not proved.
Quartic report, R1: smaller order ten	The complete all-b statement for $b \geq 6$ is proved in Theorem 5.1; the new missing certificate is $b = 962$. The smaller-order-twelve and unrestricted unequal-order questions remain open here.
Harmonic-gap continuation, Q4 and Q5	The first two-marker target and all identical marked clusters are proved by (6.17) and (6.15). Corollary 6.5 treats the fully symmetric unequal-marker component. General individual ordered unequal-marker reductions remain open.
Higher-defect report, face and odd-weight questions	Theorems 7.3–7.5 give the ordinary face decomposition and its full rational zero-jet term. The periodic extension is Corollary 7.7. Special-function evaluation of general higher face integrals remains open.

The general all-power generator is not the already known finite-span theorem for powers of the ordinary harmonic number H_n. Here the objects are powers of the decaying polygamma tail $\zeta(2, n+1)$. Likewise the face theorem is not a new claim that a Dirichlet mixture exists: it is the explicit decomposition and integrability of its boundary-subtracted pieces. The relative Hurwitz interpolation itself is inherited from published work. These separations prevent duplicate claims during integration.

8.2 Corrections and normalization details

No new false theorem was found in the targeted rank and harmonic-gap material reviewed for this continuation. In particular, the source's single-marker theorem correctly left the several-marker extension open. The lower-index-zero binomial omission in the reconstruction matrices was already explicit in the canonical fifth-order report and in the incoming extensions. It must be retained as an existing convention, not reported as a newly discovered erratum.

The following details are essential when the new results are inserted. They are precise safeguards for the supplied proof chain.

1. In the centered finite-cutoff argument use $n = 0, \dots, N-1$. The even-power subtraction sums then have zero constant coefficient. Changing to a cutoff ending at N requires its endpoint polynomial. The missing $n = 0$ endpoint also produces the exceptional $-d\delta_{k,2}\zeta(2)^{d-1}/2$ in (2.14).

2. The tail expansions are asymptotic. Only a finite subtraction, with a normally convergent remainder, is used at any specified spectral point. A fixed infinite asymptotic series is not summed as an ordinary power series in $1/n$.

3. The words "eight coordinates," "eleven coordinates," and "rank $2b$" concern the indicated rational coefficient maps. They do not assert that the corresponding real or complex constants are linearly independent over $\mathbb{Q}$. The source's existing formal rank convention is preserved.

4. In the multiple-mark formula differentiate independent parameters before specialization. A

total derivative along the diagonal is a different expression. For unequal marks, division of the symmetric sum by $k!$ is not valid. For identical marked pairs it is valid.

5. The Gamma expressions use differences of analytic log-Gamma branches on the right half-plane. An independently chosen logarithm of a quotient need not have the same branch. The Stieltjes primitive vanishes at its stated lower endpoint.

6. In the spectral decomposition the zero-jet correction $\mathcal{L}_W$ is only the contribution of $F_0(B)$. Lower derivatives $F_1, F_2, \dots$ can still contain the Stieltjes and residue terms present in the existing one-mass formulas. Removing $\mathcal{L}_W$ does not remove all such terms. The pair term $\zeta(2)$ in (7.21) is required by the Gamma-ratio normalization.

7. A simplex face term is not an irreducible connected cumulant. It can be nonzero when only two of several masses coincide. Formula (7.24) gives its precise quadratic behavior and prevents an incorrect stronger collision claim.

An independent audit strengthened the proof of the polylogarithm generator by replacing generic endpoint-regularization language with an explicit convergent-MZV formula for the sole leading-one endpoint. The Laurent derivative estimate was also written with its required logarithmic factor. These are corrections made to the present draft during review, not errors attributed to the baseline.

8.3 What the reproducible checks establish

The package separates proof certificates from numerical diagnostics. The essential exact checks are as follows.

- A finite Faulhaber calculation, independent of the new Bernoulli recurrence, agrees with it in 220 rational checks across powers one through eight, including nonpositive anchors and boundary parity. It retains MZVs as formal words. A separate finite-cutoff replay proves the reduction of (3.15) using only four classical MZV relations. Rational coefficient extraction reproduces the full eight-by-eight and eleven-by-eleven determinants and elimination rows. For the repeated even-order extension, an independent standard-library program checks 16 cubic cutoff polynomial identities and 192 finite tail-difference versus strict-composition anchor identities.

- The order-ten replay reconstructs the inherited determinant polynomial from 120 signed permutation products. It checks all 10,403 integer monomials, the Bernoulli modular substitution, and the small base matrices. A separate program computes all 25 coefficient norms used in the eventual argument and verifies the integer factorial comparison. The new exceptional certificate uses the displayed five-by-five matrix and independently computed Bernoulli residues. No numerical determinant decides nonvanishing.

- The marked-gap replay checks the formal insertion and partition algebra through five independent labels. At five labels there are 52 set partitions and 120 surviving permutations; all unwanted endpoint factors and shorter words cancel. The analytic insertion theorem is proved in the text; the program checks its finite combinatorial consequences.

Independent numerical representations then test the conclusions. Three ordinary fifth- and sixth-power identities agree to more than 90 decimal digits using finite direct sums and centered Hurwitz tail acceleration at two different cutoffs. The noninteger-endpoint marked gap and its actual zero-marker derivative are reconstructed from ordinary nested sums and compared with the decorated formulas; the latter residual at cutoff 160 is approximately 3.73×10^{-31}. Finite-interval Gamma and Stieltjes specializations agree to the stated working precision. For

the three-mass spectral third derivative, a direct spectral evaluation agrees with independent Gamma face quadrature to approximately 8.2×10^{-32} at the highest reported quadrature order.

These errors are observed cross-representation residuals. They are not interval enclosures, and common truncation errors can survive a comparison. The theorems rest on the analytic proofs and finite exact identities. The package records the working precision, cutoff, method, and exact-versus-numerical status of the checks. Its scripts use local files only and need no repository checkout or network access to run.

The archive includes a standalone LaTeX source, the compiled PDF, modular section sources, Python verification programs, successful output records, a claim-status ledger, source hashes, integration notes, and a manifest verifier. The standalone source is generated from the modular article, so the readable PDF and reusable section fragments share the same mathematics. The source repository is left unchanged.

9 Further research questions and exact next targets

The most useful next questions ask for identities, finite reductions, or proofs of specified families. The following targets arise directly from the present theorems.

R1. Determine the smallest formal coordinate system at every tail power. The universal generator supplies a finite span and exact ranks eight and eleven for the chosen fifth- and sixth-power systems. Construct canonical reduced coordinates at power d, state the admitted MZV relations, and prove the rank of the resulting moment coefficient map. One should distinguish the rank of that formal map from the unknown arithmetic dimension after evaluating its coordinates.

R2. Classify compact polynomial annihilators. Equations (3.15) and (3.21) cancel all new high-weight anchors in selected moment combinations. Find a structural description of the polynomial weights producing a prescribed lower coordinate span. A useful theorem would give the least degree and a closed basis of such weights, with their exact asymptotic subtraction polynomials. This could explain why the relatively small coefficients in (1.1) exist.

R3. Reduce the new anchored generators to explicit polylogarithmic forms. The generator theorem is a finite algorithm with a depth upper bound. Write a compact explicit expression for H_5 or H_6, then determine which identities among allowed arguments and products reduce it. Theorem statements should specify the function class and branches. A proof of minimal depth would require an appropriate functional invariant and is not supplied by the finite coordinate determinants.

R4. Find arithmetic recurrences for the resonant Laurent family. Theorem 4.1 fixes every coefficient and its spectral germ. Determine whether the dependence on r can be described by an anchored recurrence with a controlled set of new logarithmic constants. The first concrete case is the quartic constant term $\Lambda_{4,r,0}$. Any proposed recurrence must reproduce the $h_{4,r}\gamma_0(1/2)$ contribution exactly.

R5. Complete smaller order twelve, then remove the fixed-order restriction. The order-ten proof shows how a modular certificate and a proved eventual range can complement each other. Carry out the six-dimensional contraction for smaller order twelve, account for every exceptional residue below a sufficient threshold, and provide finite exact certificates. A uniform determinant identity preventing cancellation for all unequal even orders would settle the broader optimal-prefix conjecture. The full gap-six family is still an intermediate target.

R6. Find closed reconstruction rows. The ordinary identities (5.15) are explicit finite rational algorithms involving an inverse matrix. Seek a fixed-size kernel or product formula for its rows, including all endpoint constants. This is a stronger goal than nonsingularity and may yield short identities at large orders without constructing a large inverse.

R7. Resolve the antisymmetric two-marker component. For unequal marked pairs, the partition theorem gives the symmetric combination. Determine an independent expression for $\mathcal{H}(s,u,v,t)-\mathcal{H}(s,v,u,t)$ and its mixed derivatives. A proposed reduction should retain the given relative quotient at noninteger endpoints and prove an ordinary convergence domain before continuation. This is the precise part of the several-marker question not supplied by a classical symmetric-sum law.

R8. Combine marked clusters with nonpositive deletions. Unmarked deletions can shift distinct marked centers differently. Classify exactly when such operations preserve the identical-cluster or symmetric span, and identify the additional ordered coordinates when they do not. The natural first test is one nonpositive integer letter adjacent to a two-marker cluster. Independent differentiation must precede every diagonal specialization.

R9. Evaluate special Gamma face integrals. For rational shifts and selected algebraic masses, seek identities expressing $I_S(F_1)$ in a specified polylogarithmic, multiple-Gamma, or zeta-derivative class. The quadratic identity (7.24) gives a normalization check for every face size. A finite exact integral is already available at arbitrary masses; a special-function reduction would be an additional theorem.

R10. Give higher collision coefficients face by face. The quadratic coefficient in (7.25) is known for every k. Derive closed symmetric-polynomial expressions for $I_S(B^h)$ for general h, and for the logarithmically weighted integrals in the higher derivative formula. Determine a generating function that remains regular when any subset of masses coalesces. This asks for explicit identities rather than only estimates.

R11. Allow independent spectral orders and arithmetic twists. Replace the common s by independent s_i in the product spectrum. Derive the face decomposition for the resulting nonsymmetric Dirichlet parameters, then compute prescribed transverse derivatives at their joint origin. For periodic weights, project the resulting Gamma and Stieltjes formulas onto Dirichlet characters while retaining the full logarithmic normalization in the period. The mean-zero cancellation provides an exact test at every polynomial weight.

R12. Return to the Gaussian period targets through an explicit map. The new higher-depth polylogarithmic generators may supply functional identities outside previously explored relation spaces. Before claiming progress on S_6 or revised S_8, construct a precise map from one such identity to the canonical residual, including its lower-weight products and branch constants. A formal obstruction in a chosen relation space and a high-precision small residual answer different questions; neither alone proves or disproves the period identity.

R13. Formalize the finite-to-analytic interfaces. The Bernoulli boundary polynomial, finite summation identities, modular determinant certificate, set-partition inversion, and face Möbius inversion are suitable finite formalization targets. Separately formalize normal convergence of the subtracted spectral families and agreement on their defining half-planes. Validated interval quadrature would strengthen the computational diagnostics, but remains distinct from formal proof of the exact identities.

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